

7. LLM-powered Conversational Search

In this lesson

Learn about IBM watsonx Assistant's large language model (LLM)-powered conversational search: what it is, how it works, how to set it up, and how to use it.

Prerequisites

This section does not have to be done in sequence with the other lab sections. You may either use the watsonx Assistant instance from the previous sections of this lab or create a new one. This lesson will provision Assistant and Discovery services from the IBM Technology Zone (TechZone), so that you don't have to start a Watson Discovery trial.

Table of Contents

7. LLM-powered Conversational Search	1
What is conversational search?	2
Process	2
Architecture	4
LLMs in conversational search	5
Demonstrating conversational search	6
Production conversational search	7
Setting up conversational search in your virtual assistant	8
Provision the watsonx Assistant and Watson Discovery services on TechZone	8
Provision Watson Machine Learning and Object Storage services on IBM Cloud	9
Create a watsonx project	12
Load a document into Watson Discovery	16
Set up the assistant (brand new instance)	23
Set up the assistant (existing instance)	27
Add the Watson Discovery extension	29
Add the watsonx custom extension	35
Upload and configure the watsonx actions	40
Try it out!	44
Copy the new functionality to your Lendyr assistant	46
Report an issue	51

Note: Always be sure you are working with the [latest version](#) of this lab guide because the guidance in this lab will undergo changes as the integration between watsonx, watsonx Assistant, and LLMs evolves in the second half of 2023, and IBM releases its purpose-built LLM. The lab expands on this subject in greater detail.

What is conversational search?

Welcome back to the watsonx Assistant hands-on lab! The previous section of this lab, 6. *Phone and Speech*, discussed various types of integrations specific to the phone channel. This section will focus on conversational search, an LLM-powered feature that allows your virtual assistant to answer questions conversationally on a wide range of topics.

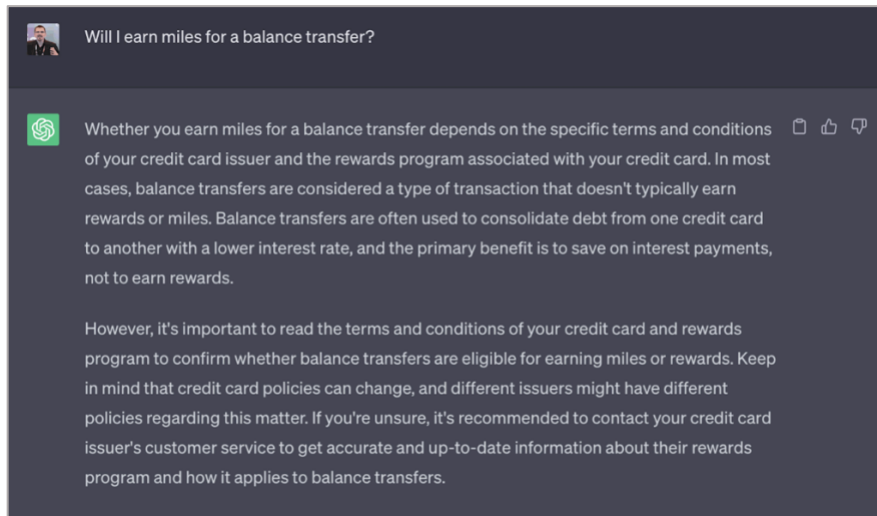
Process

Conversational search is a feature in watsonx Assistant that allows a virtual assistant to search a knowledge base for information relevant to a question and use the relevant information to generate a conversational answer to the question. This feature is an adaptation of retrieval augmented generation (RAG), a standard technical approach to improving a LLM's ability to answer questions accurately.

When it comes to enterprise use cases, where specialized and customer-specific information is involved, general purpose language models fall short. Fine-tuning these models with new corpora can be expensive and time-consuming. The lab uses the RAG technique to address this challenge.

Conversational search involves several components: conversational AI, search, and a fine-tuned or prompt-engineered LLM.

Everyone's heard of ChatGPT, and you can ask it anything – for example, *will I earn miles for a balance transfer*. It queries its large language model and generates a response:



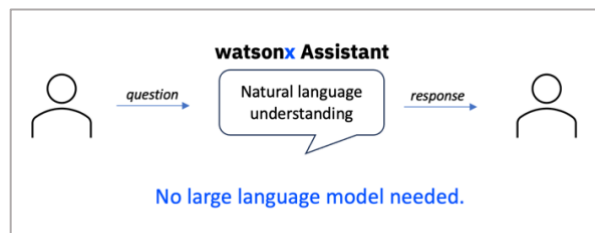
The answer is generic and demonstrates the shortcomings of the typical LLM.

In this exercise, watsonx Assistant, by integrating with watsonx, will search a knowledge base for information to find an answer to a question and use that relevant information to generate a conversational answer to the query. As mentioned above, this is an adaptation of RAG.

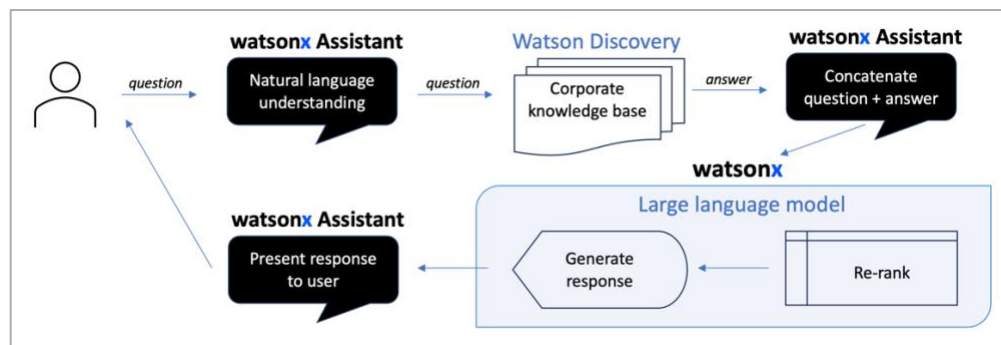
Let's walk through the order of operations in conversational search:

1. First, the end user asks the virtual assistant a question.

- The virtual assistant uses its natural language understanding (NLU) model to determine whether it recognizes the question and whether it can answer it using one of its actions for which it has been trained. For example, a bank's assistant is usually trained on answering a question such as, help me locate the nearest branch.
- If the virtual assistant recognizes the question, it answers it using the appropriate action (see lab 2. *Actions and basic IAM*). Conversational search is not needed, and watsonx Assistant did not need to call watsonx. This is illustrated in the graphic below and is how the previous sections of this lab have handled user requests.



- However, If the assistant does not recognize the question, it will go to conversational search. The conversational search process is shown and explained in detail below.



2. With conversational search, the virtual assistant sends the end user's question, also known as a query or request, to a search tool, in this exercise Watson Discovery. Watson Discovery has read and processed all relevant corporate documents.
3. The search tool (Watson Discovery) will then search its content and produce search results in response to the question.
4. The search tool passes these search results back to the virtual assistant in a list.
5. At this point, watsonx Assistant could display the results back to the user. However, they would not resemble natural speech; they would look like a summarized answer based on a

search. Helpful – certainly – but not natural.

6. Therefore, watsonx Assistant sends the question and the list of search results to watsonx, which invokes a large language model (LLM).

In some implementations of conversational search, a LLM *re-ranks* the search results. It may reorder or disregard some of the search results according to how relevant and useful it thinks the search results are to the question. For example, a LLM might decrease the ranking of a search result if it is from a document that has not been updated recently, indicating the information may be out of date. This capability is often called a “neural re-ranker.”

7. The LLM generates an answer to the question using the information in the search results, and the LLM passes this answer back to watsonx Assistant.
8. The virtual assistant presents this conversational answer to the end user.

Answer traceability. In some implementations of conversational search, the virtual assistant may show the end user the search results that the LLM used to generate the answer. This allows the end user to trace the answer back to its source and confirm for themselves its accuracy. This capability is often called “answer traceability.”

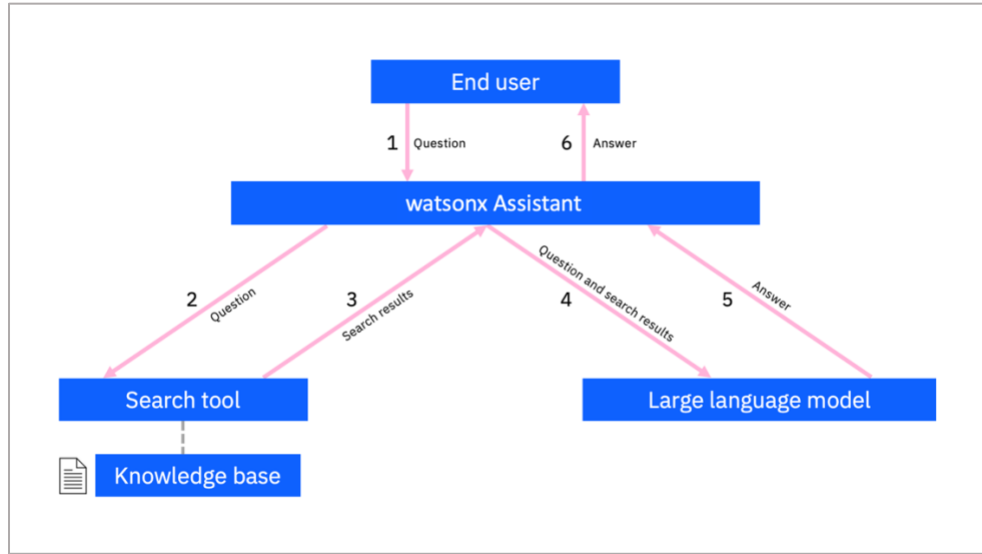
Custom passage curation. In other implementations of conversational search, the virtual assistant also shows the end user snippets of the search results that the LLM used to generate the answer. These snippets might be direct quotes or 1-2 sentence summaries of the relevant information in each search result. This allows the end user to understand exactly what the LLM pulled out from the search result to craft its answer. This capability is often called “custom passage curation.”

Conversational search is much more than document summarization or simple search. It includes the entire process of recognition, search, and answer generation; sometimes also including a neural re-ranker, answer traceability, and custom passage curation.

The **value-add** of watsonx Assistant in conversational search is its ability to orchestrate and connect every component of conversational search using its NLU model, no-code actions, OOTB connectors, and custom extensions.

Architecture

Here is a generic architecture diagram for conversational search:



LLMs in conversational search

Now that we've established that this process uses LLMs – what specific LLMs are used in conversational search?

IBM watsonx Assistant uses large language models trained and deployed in watsonx for a variety of use cases, including conversational search. Assistant customizes watsonx LLMs for conversational search such that, given a question and a list of search results, they know how to re-rank the search results and then generate an answer from that list of search results.

The watsonx Assistant team is working with IBM Research and the watsonx team to develop a custom, fine-tuned, and prompt-engineered large language model called **Content-Grounded Assistant (CoGA)** that specializes in generating answers from search results. CoGA is available in watsonx tech preview today. It will be available in watsonx later in Q3 2023.

In the meantime, while the watsonx Assistant team works to make CoGA available on watsonx, the watsonx Assistant team recommends using prompt-engineered FLAN-UL2-20b, an open-source model, to generate answers from search results. FLAN-UL2-20b is available in watsonx today.

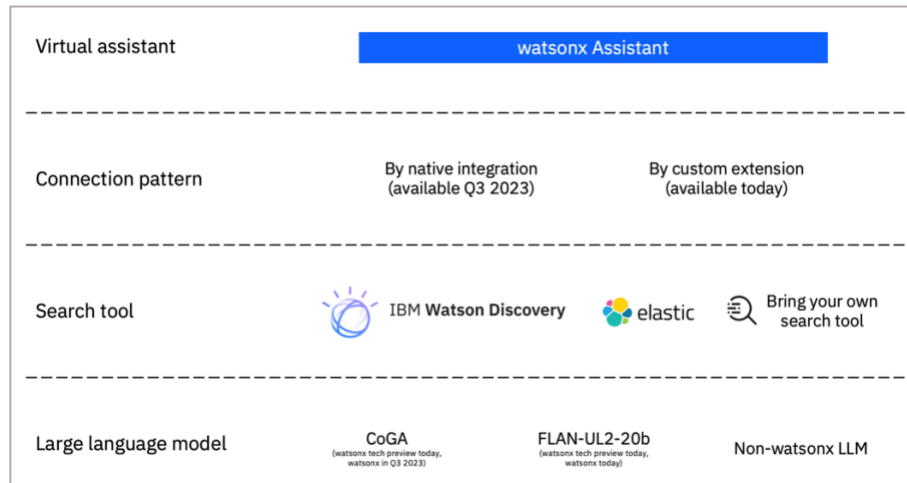
Non-watsonx LLMs can also be fine-tuned or prompt-engineered to perform conversational search. These LLMs can be integrated with watsonx Assistant via extensions. This pattern may be preferable to clients who train their own LLMs in-house.

	CoGA	FLAN-UL2-20b
watsonx <i>tech preview</i>	Available today	Available today
watsonx	Available later in Q3 2023	Available today

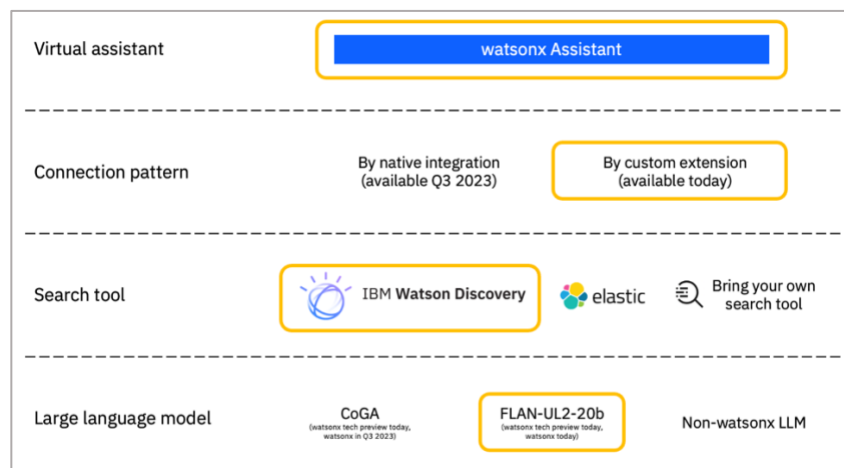
The **value-add** of watsonx in conversational search is its customized large language model, especially CoGA, designed and customized by the watsonx Assistant, IBM Research, and watsonx teams to perform well for conversational search.

Demonstrating conversational search

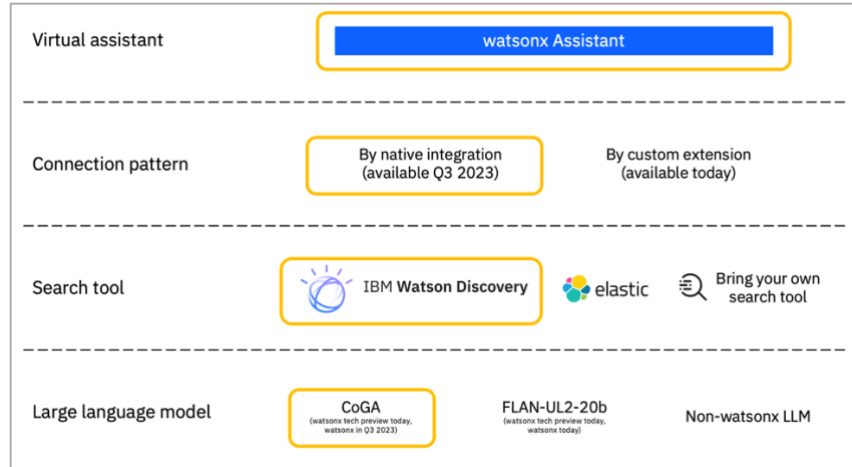
Following are the options available to IBMers and IBM business partners that would like to demo conversational search:



For a demo **today, if you do not have access to watsonx tech preview**, IBM recommends watsonx Assistant, by custom extension, Watson Discovery, and FLAN-UL2-20b. This section of the lab implements this scenario, as emphasized below.



Later in Q3 2023, once CoGA is available on watsonx and available in watsonx Assistant natively, IBM will update the recommendation to be watsonx Assistant, by native integration, to Watson Discovery, and CoGA:



Production conversational search

For production **today**, IBM recommends watsonx Assistant, by custom extension, Watson Discovery, and FLAN-UL2-20b; or watsonx Assistant with NeuralSeek using watsonx LLMs.

Later in Q3 2023, once CoGA is available on watsonx and available in watsonx Assistant natively, IBM will update the recommendation to watsonx Assistant, by native integration, Watson Discovery, and CoGA.

Setting up conversational search in your virtual assistant

Let's walk through how to set up conversational search using watsonx Assistant, by custom extension, with FLAN-UL2-20b.

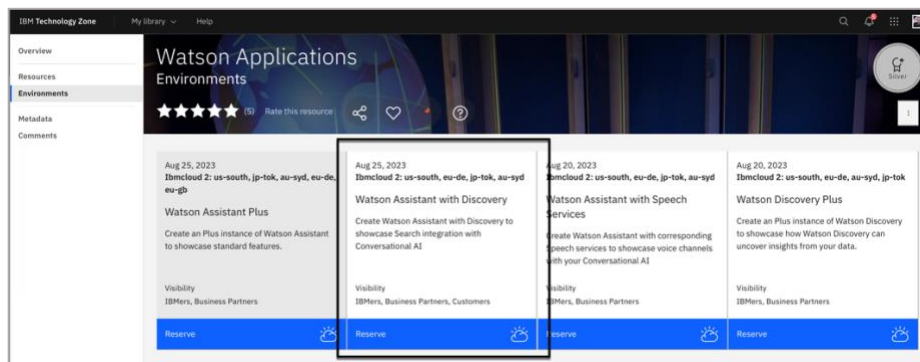
In this lab section, you will create a new virtual assistant. At the end of this exercise, you will have the ability to transfer the capabilities you will build in this section, to the assistant you built in Sections 1-6, but only if both assistants are running within the same instance of watsonx Assistant. You will [copy-and-paste](#) the actions and reconfigure the extensions.

For a preview and demonstration of what you are about to build in this section, view [this video](#).

Provision the watsonx Assistant and Watson Discovery services on TechZone

Important: If you have just completed Section 6 of this lab series and still have your TechZone environment running, skip this part and go to the header, *Provision Watson Machine Learning and Object Storage services on IBM Cloud*.

Navigate to the [Watson Applications](#) collection on TechZone, and click on the **watsonx Assistant with Discovery** tile to reserve an environment:



This will take you to the reservations page.

1. Enter a **name** for the reservation that is meaningful to you.
2. Under **Purpose**, select the **Practice / Self-Education** tile.
3. Enter a **Purpose description**.
4. Select a **Preferred geography**.
5. You can leave **End date and time** as-is, as it defaults at the maximum 2 days.
6. Accept the **Terms & Conditions**.
7. Click **Submit**.

IBM Technology Zone My library Help

Create a reservation

Watson Applications [View collection](#)

Select an environment/infrastructure Select a reservation type Fill out your reservation Complete

Name
Watson Assistant with Discovery 1
Name this reservation. This will help identify it in your reservation list.

Purpose
Customer Demo
Showing a pre-built, standardized product use case with minimal customization needed on the environment for a client demonstration. Aligns with "Custom Demo" in IBM Sales Cloud.
Practice / Self-Education 2
Gaining experience with specific technology, product, or solution.
Proof-of-Technology
Customizing an environment to set up and personalize beyond the pre-built option to showcase a specific customer use case. Typically these environments are needed for a longer period of time. Aligns with "Proof of Value" in IBM Sales Cloud.
Test
Need to test a specific function, configuration, or customization.

Please ensure to select the correct purpose as this can NOT be updated or changed after this reservation has been created. Review the [Reservation Duration Policy](#) to understand default durations allowed for specific infrastructures based on purpose.

Sales Opportunity number
Enter an opportunity number
Providing an [IBM Sales Cloud prospectus number](#), [Salesperson Relationship ID](#), or a [Project Work ID](#) will allow you to extend your reservation date.

Purpose description
Part of my Your Learning course 3

What are you doing? Why do you need that? What are you trying to accomplish?

Preferred Geography
AMERICAS - us-south region - dal10 datacenter 4

End date and time
Select a date
08/23/2023
Select a time
7:40 PM 5
America/Chicago

Reservation policy: Recommended 2 days, but can be reserved up to 2 days on the reservation form. Extend later for 2 days increments up to 4 days total. Max time 6 days total.

6 I agree to IBM Technology Zone's [Terms & Conditions](#) and [End User Security Policies](#)

7 Submit

You will soon receive an email that your environment is provisioning. While you wait for it to be provisioned (usually less than 10 minutes), you can continue with the next steps.

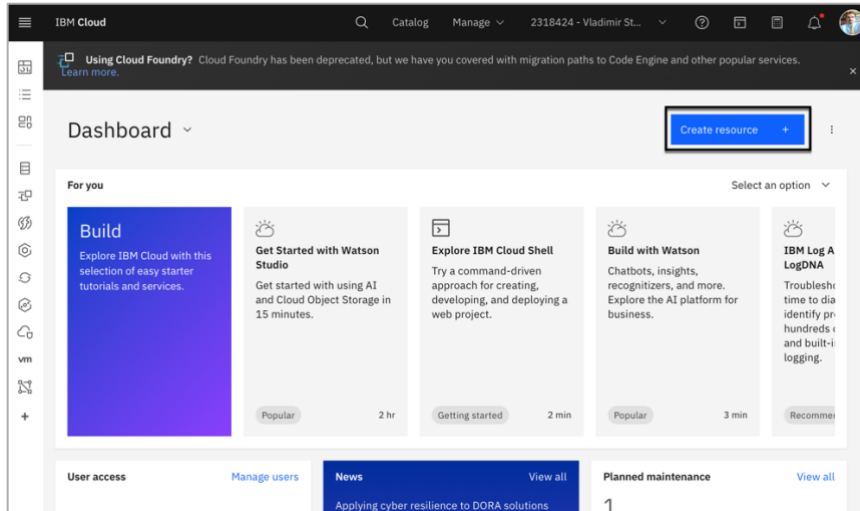
Note: This environment can be extended up to two times, for a maximum of 6 days in total. You will receive multiple expiration warnings from TechZone before the environment expires, which will give you an opportunity to extend it, up to the 6-day maximum.

Provision Watson Machine Learning and Object Storage services on IBM Cloud

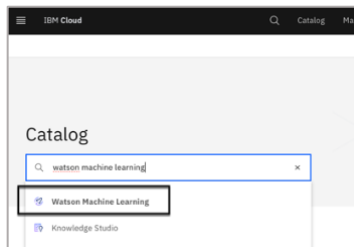
IBM watsonx will need instances of Watson Machine Learning and Object Storage to connect to and communicate with watsonx Assistant and the LLM. If you already have existing instances, you can reuse them and skip this step.

Note: IBM services typically allow you to have no more than one Lite instance of a service. You may have to delete the existing Lite service before creating another one.

First, log onto [IBM Cloud](#) and click on **Create resource +:**

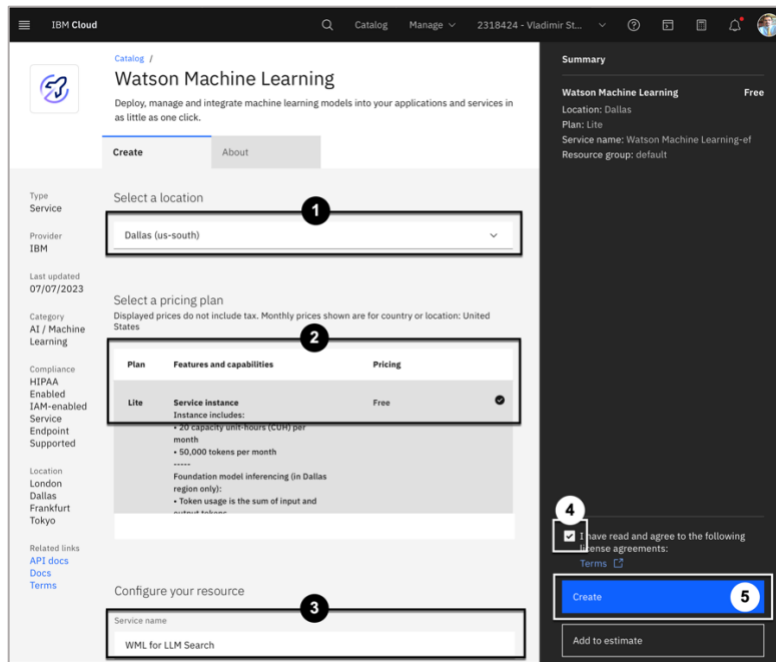


In the service catalog, search for, and select **Watson Machine Learning**:



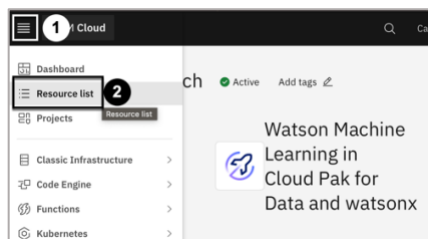
On the create service page,

1. **Select a location.**
2. Select the **Lite** pricing plan and note that it is **Free**.
3. Scroll down until you are able to enter a **Service name** that is meaningful to you.
4. Agree to the **Terms & Conditions**.
5. Click **Create**.

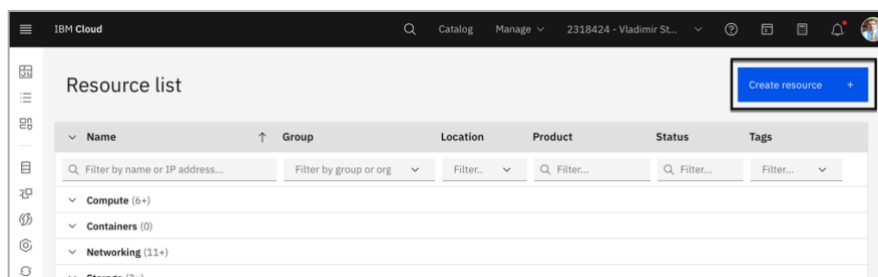


This will take you to the newly created Watson Machine Learning service page, but you won't need to do any further configuration to this service. Instead, you will now provision an Object Storage service. Watsonx will use object storage for its own metadata and processing.

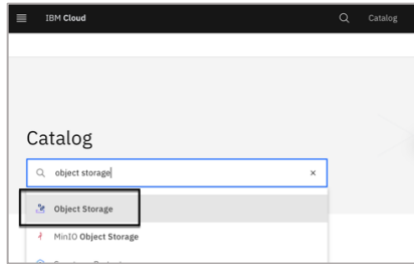
Click the IBM Cloud **menu icon (1)**, and select **Resource List (2)**:



The Resource list page is shown. Here, click **Create resource +**.

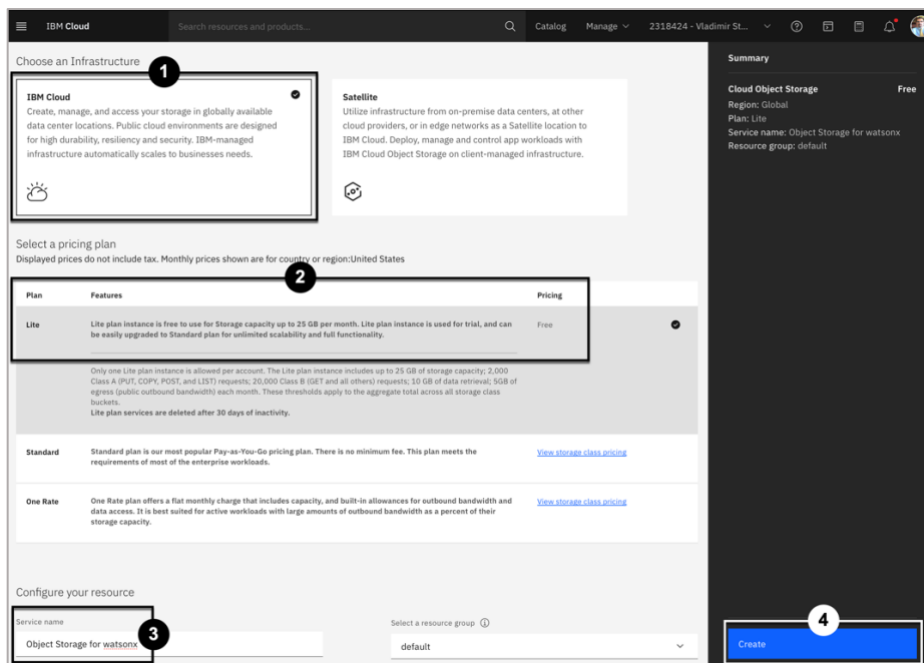


Search for, and select, **Object Storage**.



On the service creation page,

1. Select **IBM Cloud**.
2. Pick the **Lite** plan and note that it is **Free**.
3. Enter a **Service name** that is meaningful to you.
4. Click **Create**.

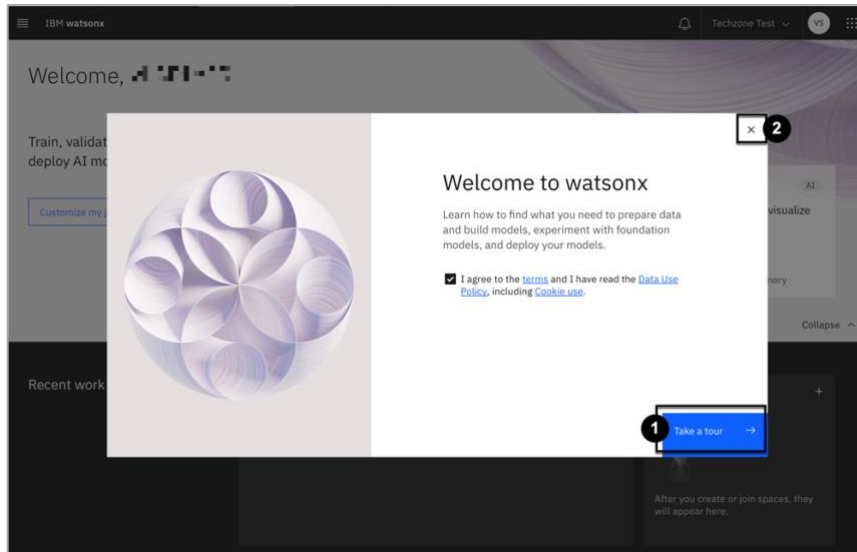


You have now provisioned watsonx Assistant and Watson Discovery through TechZone, and Watson Machine Learning and Object Storage through your IBM Cloud account. Next, you will provision a watsonx project.

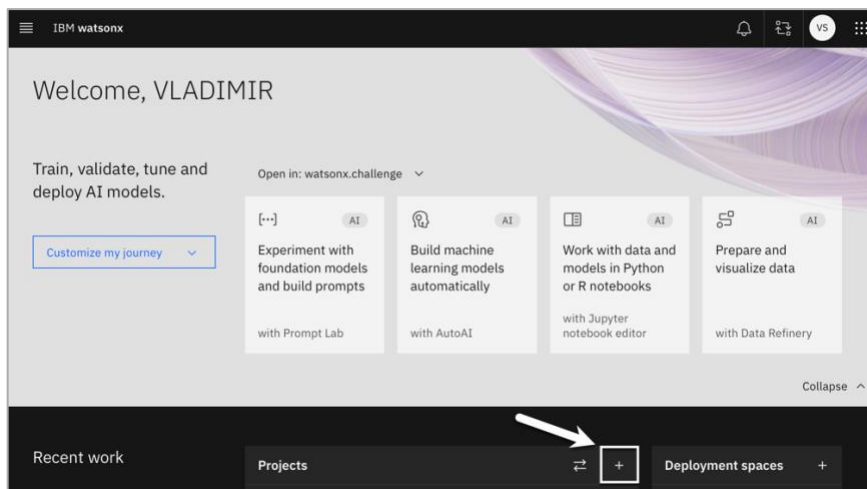
Create a watsonx project

IBM watsonx is an enterprise-ready AI and data platform designed to multiply the impact of AI across a client's business. It provides an API for interacting with generative language models. In this step, you will create a watsonx project, which will later allow you to connect watsonx Assistant to the watsonx API.

First, navigate to [watsonx](#). You may see a splash screen like below. If so, you may **Take a tour (1)** or **exit (2)** the screen.



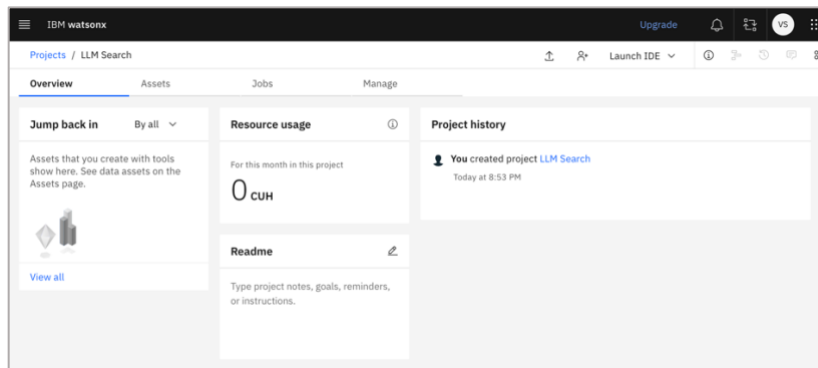
Once on the main watsonx page, create a new project by clicking the + button under **Projects**:



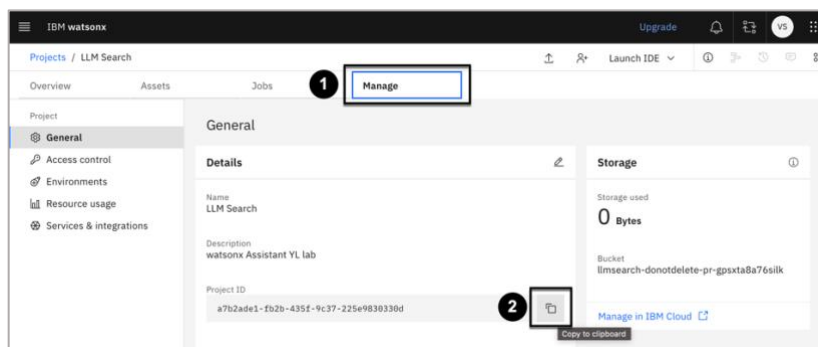
On the **New project** screen,

1. Enter a **Name** that is meaningful to you.
2. Optionally, enter a **Description**.
3. Select the **Object storage** service you created in a previous step.
4. Click **Create**.

Next, the **Overview** tab of your new watsonx project screen is shown.

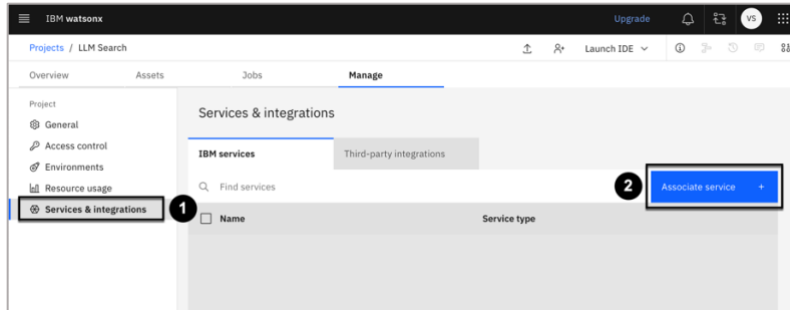


You will need the watsonx project ID to set up the action that calls the watsonx custom extension in Assistant later on. To get the watsonx project ID, select the **Manage (1)** tab, then click **copy (2)**.

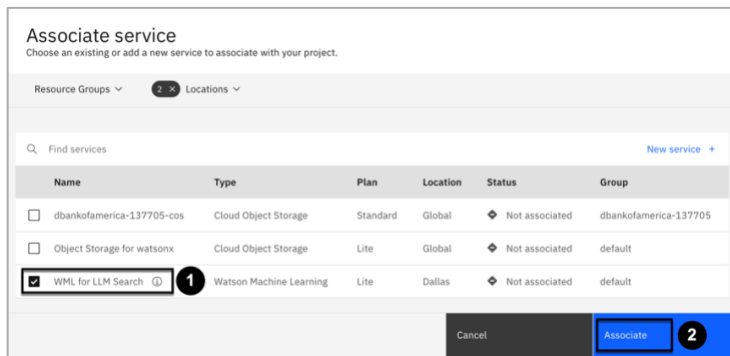


Save this project ID in a notepad for now, as you will need it shortly.

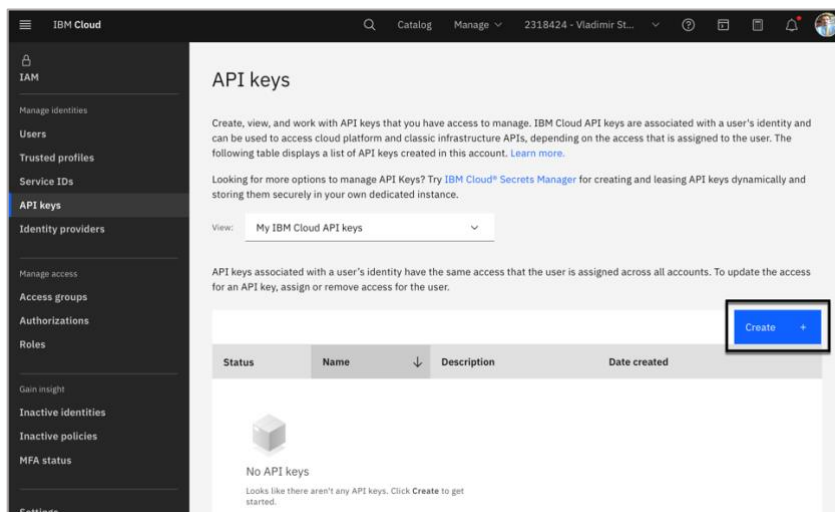
Next, you will link the Watson Machine Learning service instance you created earlier to this new watsonx project. To do this, select **Services & integrations (1)** and click **Associate service + (2)**:



On the next screen, select your **Watson Machine Learning** instance (1) and click **Associate** (2):



Setting up your watsonx extension in Assistant will also require an API key from your IBM Cloud account. To get the API key, in a new browser tab, navigate to [Manage access and users - API Keys](#) in IBM Cloud, which will take you to the following screen. Then, click **Create +**:



On the popup *Create IBM Cloud API key* screen, enter a **Name** (1) and **Description** (2) meaningful to you, then click **Create** (3):

Create IBM Cloud API key

Name: LLM search project 1

Description: for watsonx Assistant LLM search project 2

Cancel Create 3

When you see the notification API key successfully created, click **copy**:

API key successfully created

Copy the API key or click download to save it. You won't be able to see this API key again, so you can't retrieve it later. The API key is no longer displayed after 292 seconds.

API key

Copy Download

Save this API key somewhere safe and accessible. You will need this API key later to set up the watsonx custom extension in Assistant.

You can close this IBM Cloud browser tab.

Load a document into Watson Discovery

Next, you will open your Watson Discovery instance.

From your email notifying you that your TechZone environment is ready, click on **View My Reservations**.

IBM Technology Zone <noreply@techzone.ibm.com> Today at 7:57 PM
To: VLADIMIR Stojanovski

This Message is From an External Sender
This message came from outside your organization. Report Suspicious

IBM Technology Zone

Status Update: **Ready**

Your reserved environment has successfully provisioned and is ready to use. Select "View My Reservations" button below to navigate to your reservations.

Reservation Name:
Watson Assistant with Discovery

Reservation ID:
64e405779ca0c00017852b38

Start Date :
2023-08-22 00:46:00 (UTC Time)

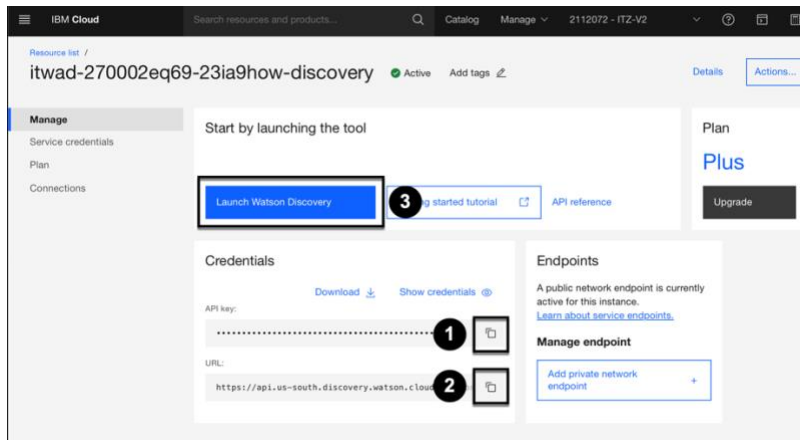
End Date :
2023-08-24 00:51:00 (UTC Time)

Collection:
<https://techzone.ibm.com/collection/63860fb337f8a600183c731e>

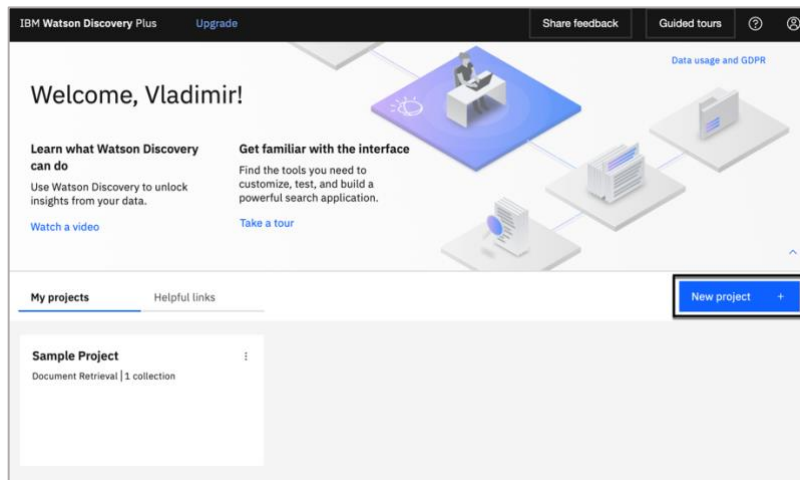
Environment Name:
ltwad-270002eq69-23ia9how

IMPORTANT: If this environment requires an access to IBM Cloud and you are not a member of DTEV2 account, you will be invited to join it. Please go [HERE](#) to accept your invitation.

[View My Reservations](#)



On the welcome page, click **New Project +**.



Enter a **project name (1)** such as “Lendyr documents” and select **Document Retrieval (2)**, then click **Next (3)**.

IBM Watson Discovery Plus Upgrade My projects Share feedback Guided tours

Select project type Select data source Connect to data Configure collection

What type of project are you working on?

1 Project name
Lendyr documents

2 Project type

Document Retrieval
Search and find the most relevant answers from your data.

Conversational Search
Supply answers to a virtual agent built with IBM Watson Assistant.

Content Mining Enterprise
Discover hidden insights, trends, and relationships in your data.
Project type only available with the Enterprise plan. Upgrade

None of the above – I'm working on a custom project

Working with contracts? Enterprise
Let us do the heavy lifting. Watson Discovery can automatically classify parties, amendments.

3 Next →

Download the [Lendyr FAQ document](#). This is the file you will upload to Watson Discovery. Once downloaded, click **Upload data (1)** hit **Next (2)**.

IBM Watson Discovery Plus Upgrade My projects Share feedback Guided tours

Select project type Select data source Configure collection

OK, where is your data?

Salesforce

SharePoint Online

SharePoint On Prem

Box

Web crawl

IBM Cloud Object Storage

1 Upload data

Reuse data from an existing collection

Don't see the data source you need?
Let us know

2 Next →

Enter a **collection name (1)** such as “Lendyr documents” and click **Next (2)**.

IBM Watson Discovery Plus Upgrade My projects Share feedback Guided tours

Select project type Select data source **Configure collection**

Let's create a collection for your data

General

Data source

Upload data

1 Collection name
Lendyr documents

Collection language
English

More processing settings

Back Next

Now, **upload (1)** the FAQ file you downloaded, and click **Next (2)**.

IBM Watson Discovery Plus Upgrade My projects Share feedback Guided tours

Select project type Select data source **Configure collection**

Upload complete!

Now you can upload more files or finish and continue to your new project.

Upload more files

Max file size is 10MB. Supported file types are PDF, HTML, JSON, Word, Excel, PowerPoint, PNG, TIFF, JPG, GIF, TXT, CSV, ZIP, GZIP, TAR.

1 Drag and drop files here or upload

Back Finish

The file uploads relatively quickly, however it may take 15 minutes for Watson Discovery to process it. To monitor this progress, click **Manage collections (1)** and select the **Lendyr FAQ** collection tile:

IBM Watson Discovery Plus Upgrade My projects Share feedback Guided tours

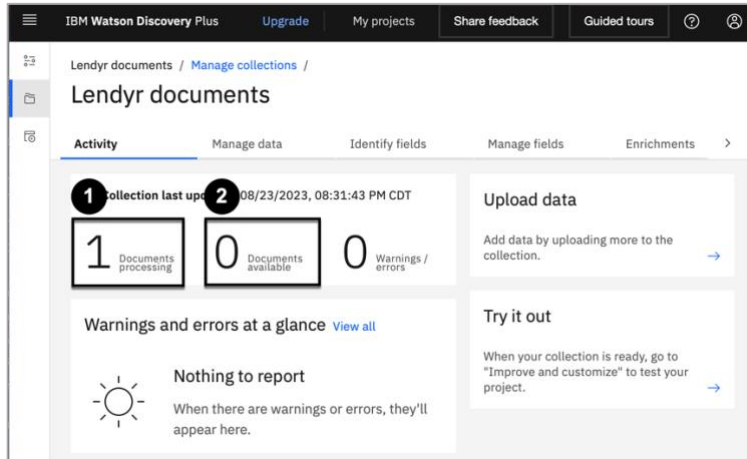
Lendyr documents / **Manage collections** New collection

1 Manage collections

2 Lendyr FAQ

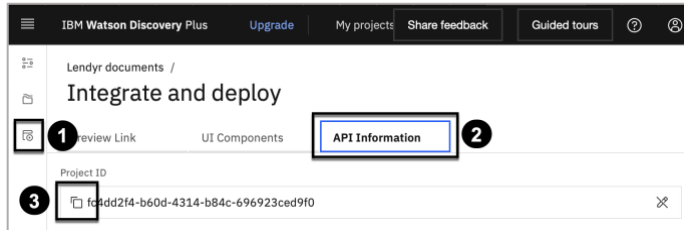
0 documents
Last updated: 8/22/2023

The next screen shows the collection state. Note that your document is still processing **(1)** and there are **0 Documents available (2)**. The processing will finish when the document becomes available.

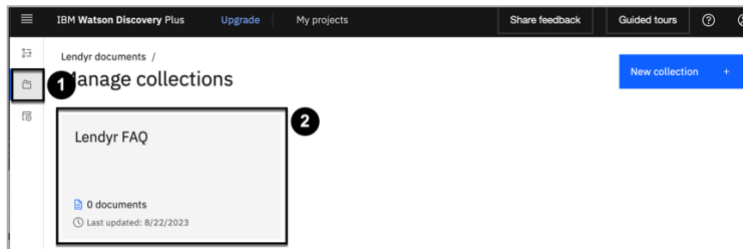


While you wait for the document to process, get the Watson Discovery project ID. You will need it when you configure the watsonx Assistant action.

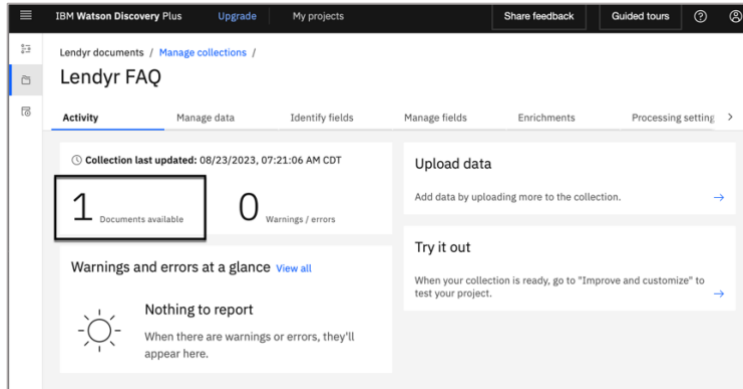
To get the project ID, select **Integrate and deploy (1)**, click the **API Information (2)** tab, and copy the **Project ID (3)**. Save it elsewhere.



Go back to the document collection, so you can continue to monitor its progress, by clicking **Manage collections (1)** and **Lendyr FAQ (2)**:

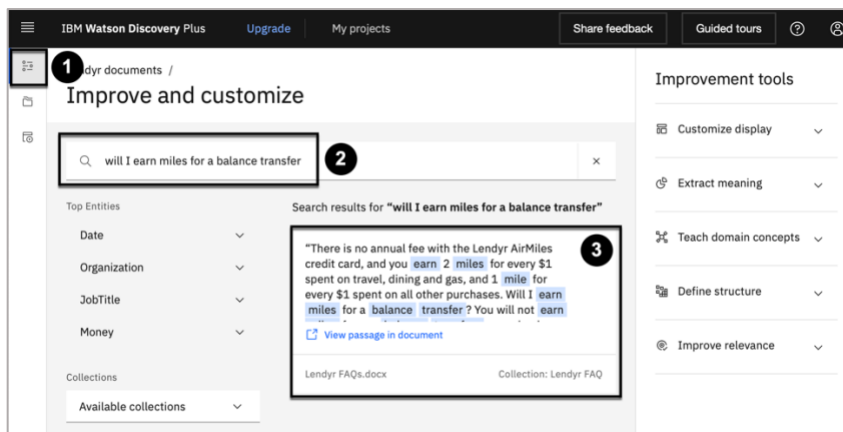


Once the screen says **1 Documents available**, you can proceed to the next step. This processing usually takes 5 to 15 minutes, so it may be a good time to take a short break.



Now that the processing is finished, let's ask Watson Discovery the same balance transfer question we asked ChatGPT at the beginning of this lab section: *will I earn miles for a balance transfer*.

To do this, click on **Improve and Customize (1)**, then type **will I earn miles for a balance transfer (2)** in the query window. Note the **search result (3)**:



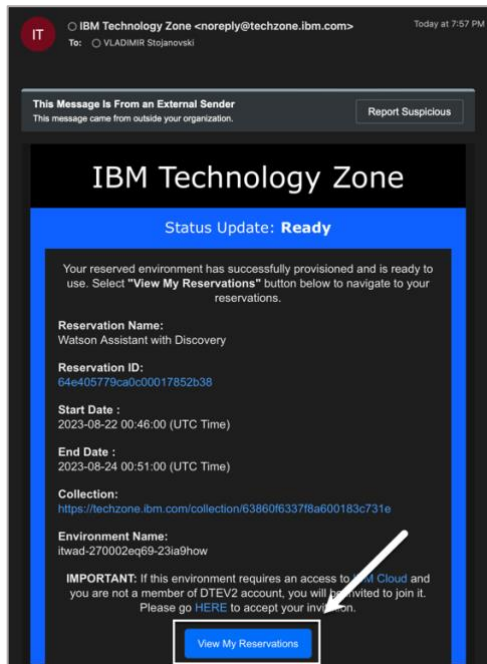
Watson Discovery provides a helpful answer, but not a conversational one; it's an excerpt from a business document that Watson Discovery read. Still, note that it is much more helpful and useful to the user ("You will not earn miles for any balance transfers ...") than ChatGPT. This is because Watson Discovery understands the business content and context of an organization.

This lab will apply the power of generative language models to produce a more conversational and succinct answer to this question.

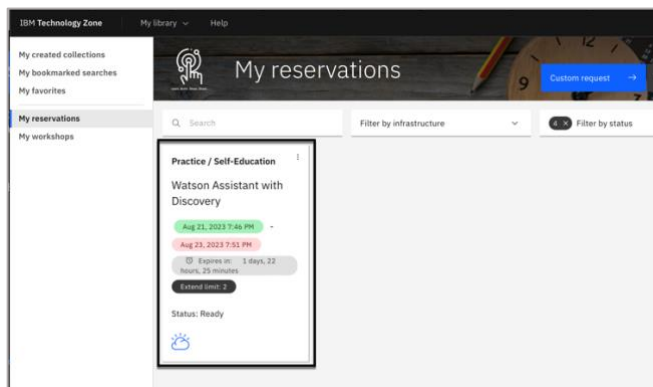
Set up the assistant (brand new instance)

Important: If you completed Section 6 of this lab series and still have your TechZone environment running, skip this part and go to the header, *Setup the assistant (existing instance)*.

Go back to the email you received from TechZone, letting you know that your environment is ready. From that email, click on **View My Reservations**.



This takes you to the **My reservations** page on TechZone. Click your instance tile:



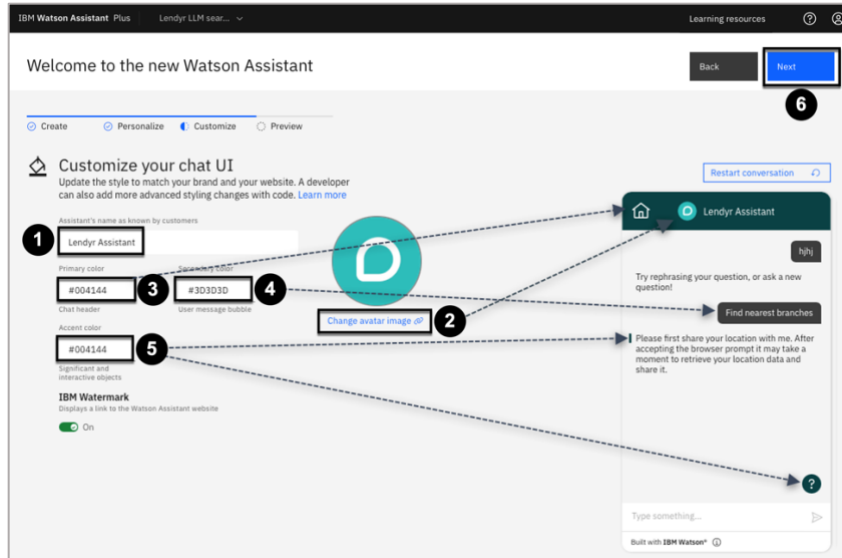
On your TechZone instance page, click the **watsonx Assistant URL**:

Fill out the form asking you how you will use the assistant **(1)** and click **Next (2)**:

Customize your assistant to look like the Lendyr Bank website

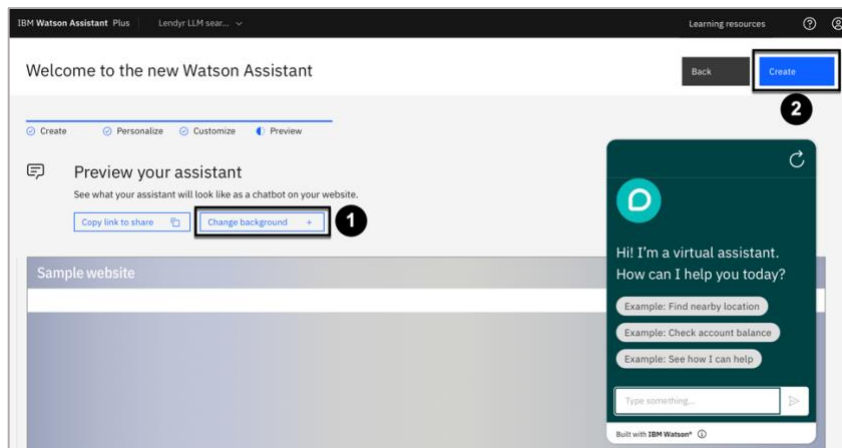
The next screen allows you to customize the look and feel of the Assistant; customize it to look like the Lendyr Bank assistant (you will recall going through these steps in Section 2 of this lab):

1. Change **Assistant's name as known by customers** to “Lendyr Assistant”.
2. Click on **Add an avatar image** and point to <https://web-chat.global.assistant.watson.appdomain.cloud/assets/Lendyr-Avatar.png>.
3. Change **Primary color** to: #004144. This is the color of the chat header.
4. Change **Secondary color** to: #3D3D3D. This colors the message bubble.
5. Change **Accent color** to: #004144. This is a tertiary color that accentuates certain assistant responses and icons, as shown below.



Your assistant should now look like the image above. (Note, some color changes may require that you restart the chat.)

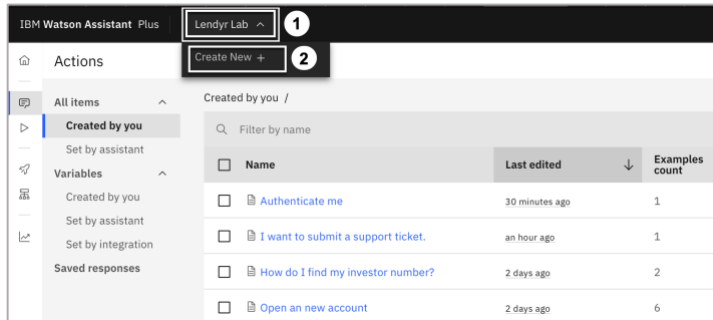
On the following preview screen, you have the option to **change the background (1)**. We won't do that because we'll embed the assistant into the Lendyr site, but this is a great option to customize the look of an assistant so it feels like it is embedded in a client's website. Click **Create (2)**.



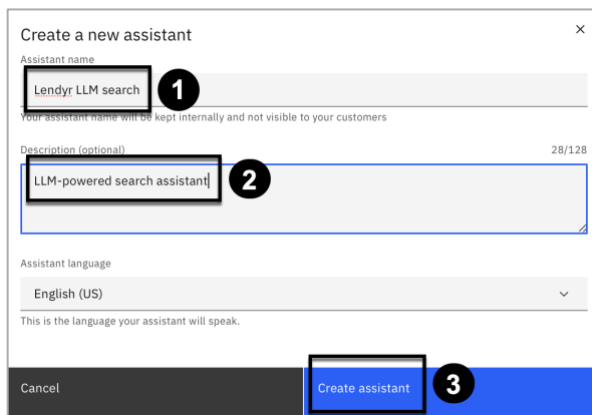
Important: Skip the next part, *Set up the assistant (existing instance)*, and move directly to the header *Add the Watson Discovery extension*.

Set up the assistant (existing instance)

In your existing Lendyr Lab assistant, click on Lendyr Lab and Create New +. You will now create a new assistant for this lab section. Later, you can move this functionality into your original Lendyr Lab assistant.



This is a new assistant, so you will first configure the basics. First, enter its **name (1)**, **description (2)**, and click **Create assistant (3)**:

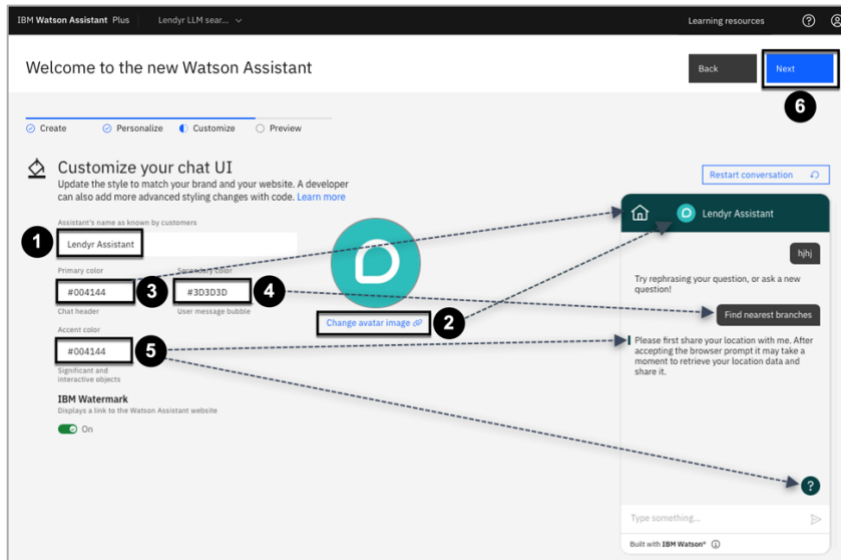


Customize the assistant to look like the Lendyr Bank website

Customize the look and feel of the Assistant to be like the Lendyr Bank assistant (you may recall going through these steps in Section 2 of this lab). If you are creating a new instance, the screen below will show automatically as part of the step-by-step wizard. If you are using an existing instance, you can find the look and feel options below by going to **Preview → Customize webchat**:

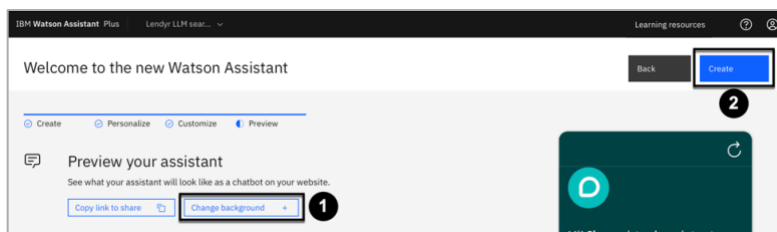
1. Change **Assistant's name as known by customers** to "Lendyr Assistant".
2. Click on **Add an avatar image** and point to <https://web-chat.global.assistant.watson.appdomain.cloud/assets/Lendyr-Avatar.png>.
3. Change **Primary color** to: #004144. This is the color of the chat header.
4. Change **Secondary color** to: #3D3D3D. This colors the message bubble.

5. Change **Accent color** to: #004144. This is a tertiary color that accentuates certain assistant responses and icons, as shown below.

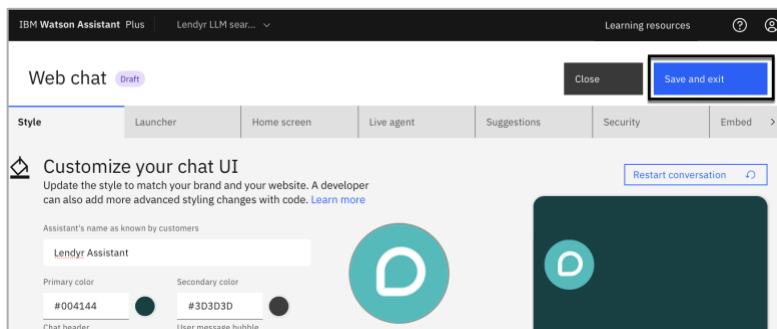


Your assistant should now look like the image above. (Note, some color changes may require that you restart the chat.)

If you are creating a new instance, you have the option to **change the background (1)**. You won't do that because you'll embed the assistant into the Lendyr site, but this is a great option to customize the look of an assistant so it feels like it is embedded in a client's website. Click **Create (2)**.



Otherwise, you are using an existing instance, click **Save and exit**:



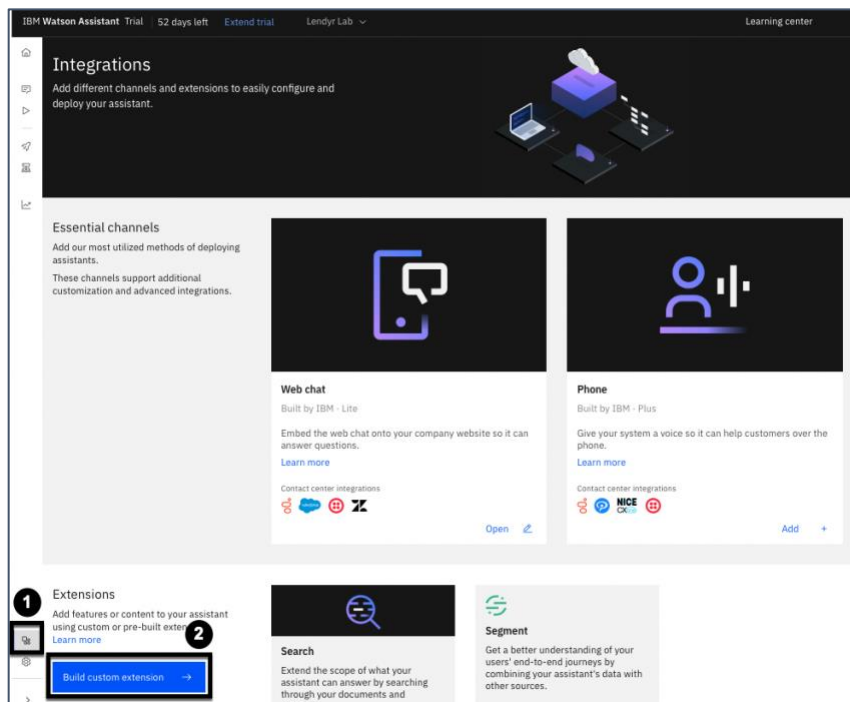
Add the Watson Discovery extension

Now that you have a Watson Discovery instance setup, a watsonx project ready, and an Assistant configured, you can proceed to add the custom extensions to your assistant.

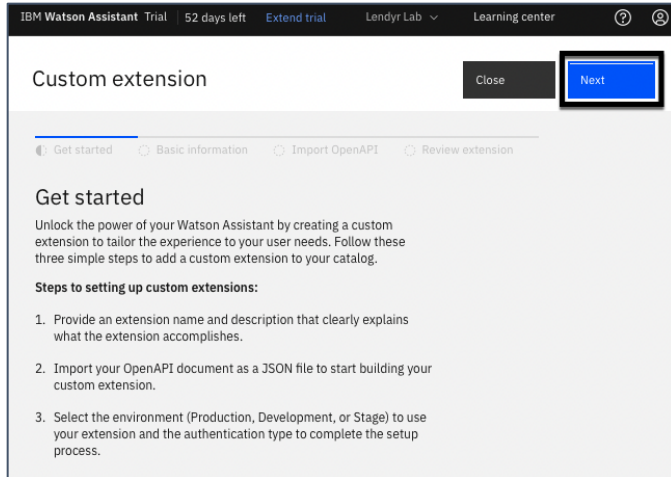
Note: Extensions were covered in detail in [Section 5](#) of this lab.

For the Watson Discovery extension, download the OpenAPI specification [watson-discovery-query-openapi.json](#). This is the API specification for the Watson Discovery custom extension.

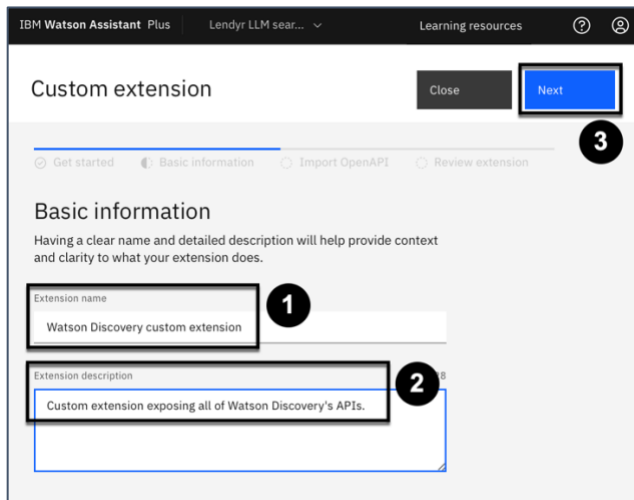
In watsonx Assistant, use the left menu to open the **Integrations (1)** page. Then, scroll down and click the **Build custom extension (2)** button:



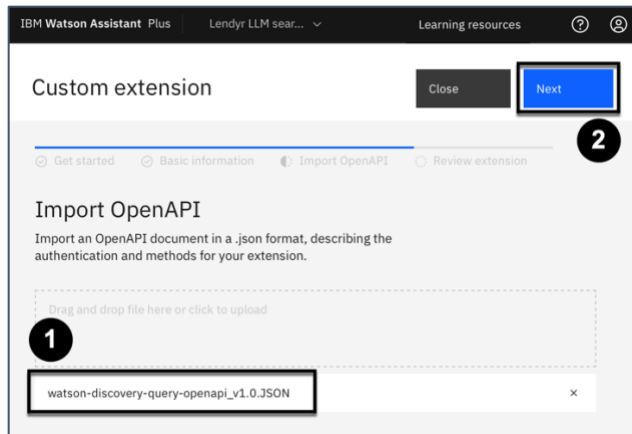
This first screen describes what you'll need to create the extension. Note that the OpenAPI JSON document, which you just obtained, is key to the setup. Click **Next** to proceed to the next screen:



The second screen asks you to name and describe the custom extension. Name the custom extension **Watson Discovery custom extension (1)** and add a description, like “Custom extension exposing all of Watson Discovery’s APIs” **(2)**. Click **Next (3)** to proceed to the next screen.

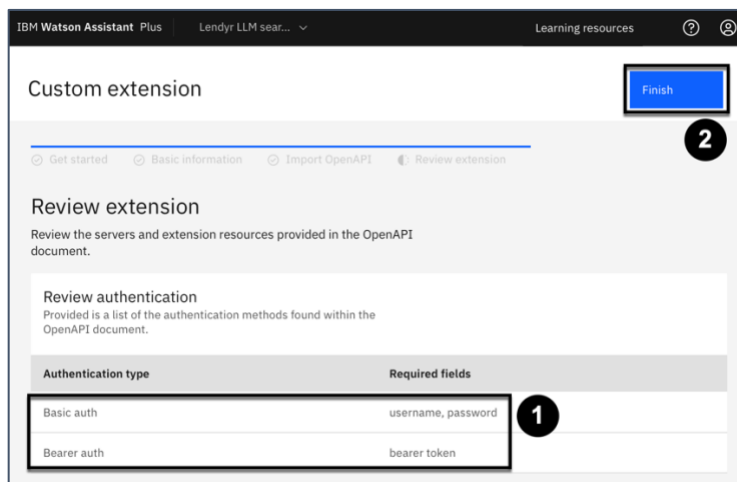


In the next screen, either drag-and-drop or click to upload the [watson-discovery-query-openapi.json](#) OpenAPI spec you downloaded **(1)**, then click **Next (2)** to proceed to the next screen:

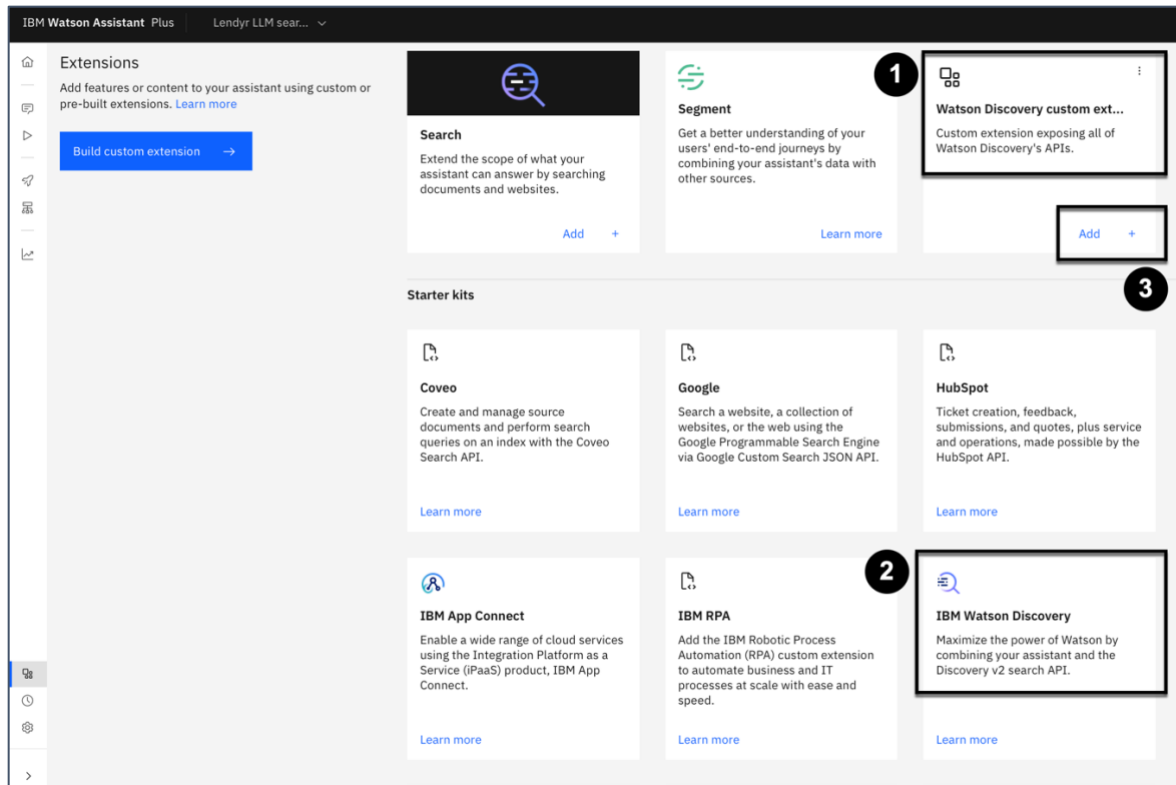


Note: If you get an error that the file is not in the proper JSON format, try downloading it again with another browser, and assure it has a **.json** extension.

Take a moment to review the extension on the following screen. This extension will allow us to call Watson Discovery by using its **API Key (1)** which you obtained earlier. Click **Finish (2)**.



You should now be able to see the **Watson Discovery custom (1)** extension in your Integrations catalog:



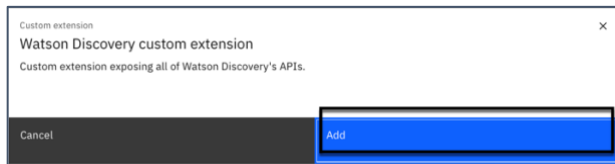
Note that it is different from the **IBM Watson Discovery (2)** extension, which is normally used as Assistant's search skill. The custom skill gives watsonx fuller access to Watson Discovery's APIs than what the search skill provides. Click **Add + (3)**.

A deeper dive on the Watson Discovery custom extension

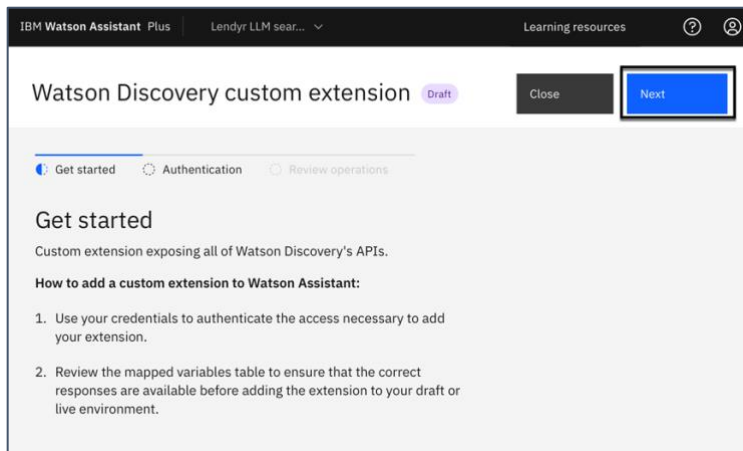
Q: Why am I not using the Watson Discovery built-in search extension?

A: An alternative to creating a custom Watson Discovery extension is to use the [built-in search integration](#) (also referred to as the *Watson Assistant search skill*). The search skill is a purpose-built integration between Assistant and Discovery for a simple, quicker, and easier experience. However, it does not expose and use all of the API parameters and the full power of Watson Discovery, which will be used by watsonx. For example, the built-in search skill has an "Emphasize the Answer" capability which allows Watson Assistant to identify the snippet of text within each response that it thinks is most likely to be the direct, concise answer to the question that was asked and will emphasize that text within the search result card. That can be a great feature when not using LLMs, but it can be detrimental when integrating with an LLM, especially if the result has no useful answer. Furthermore, Watson Discovery provides an answer *confidence* for each answer that provides an estimate of the probability that the answer is correct, so a user of Watson Discovery can use that probability to decide whether the answer is plausible enough that it should be emphasized. The built-in search integration in Watson Assistant does not expose the answer confidence because it is designed to provide simple, minimal functionality. In contrast, a custom extension lets you access all of the outputs of Watson Discovery, so that low confidence answers can be ignored.

You have created the Watson Discovery custom extension, and now you need to specify which Watson Discovery instance it will access. On the popup window, click **Add**.



On the following screen, you can read about how to add the custom extension. When ready, click **Next**:



The following screen asks for **Authentication** details:

1. Select **Basic auth** as the authentication type.
2. In the **Username** field, enter **apikey**.
3. In the **Password** field enter the **Discovery API Key** you saved earlier.
4. At the bottom of the Authentication pane, under **discovery_url**, fill in the URL for your Discovery instance which you saved earlier. Make sure you enter it without the **https://** prefix.
5. Click **Next (5)**.

IBM Watson Assistant Plus

Lendyr LLM sear...

Learning resources

Watson Discovery custom extension

Draft

Close

Next

Get started

Authentication

Review operations

Authentication

Authentication types are determined in the OpenAPI document and provide security for the extension.

Authentication type

Basic auth

Username

apikey

Password

.....

Servers

https://{discovery_url} - The Watson Discovery URL as specified in th...

Generated URL:

https://api.us-south.discovery.watson.cloud.ibm.com/instances/bb591279-a40a-4c61-9259-fa4929596a97

Server variables

Enter values for your server variables to create a valid generated URL for requests.

discovery_url

api.us-south.discovery.watson.cloud.ibm.com/instances/i

You'll see a summary view of your new custom extension. Click **Finish**:

IBM Watson Assistant Plus

Lendyr LLM sear...

Learning resources

Watson Discovery custom extension

Draft

Close

Finish

Get started

Authentication

Review operations

Review operations

This table shows the operations defined in the OpenAPI document.

Operation	Method	Resource
Query a project	POST	/v2/projects/{project_id}/query

Click **Close**:

IBM Watson Assistant Plus | Lendyr LLM sear... | Learning resources

Watson Discovery custom extension Draft Close Save and exit

Authentication | Review extension

Authentication

Authentication types are determined in the OpenAPI document and provide security for the extension.

Authentication type

Basic auth

Username

apikey

Password

.....

Servers

https://{discovery_url} - The Watson Discovery URL as specified in th...

Add the watsonx custom extension

Next, you will create the watsonx custom extension. First, download the [watsonx OpenAPI specification file](#). This JSON file defines the watsonx custom extension.

In watsonx Assistant, use the left menu to open the **Integrations (1)** page. Then, scroll down and click the **Build custom extension (2)** button:

IBM Watson Assistant Trial | 52 days left | Extend trial | Lendyr Lab | Learning center

Integrations

Add different channels and extensions to easily configure and deploy your assistant.

Essential channels

Add our most utilized methods of deploying assistants. These channels support additional customization and advanced integrations.

Web chat

Built by IBM - Lite

Embed the web chat onto your company website so it can answer questions.

[Learn more](#)

Contact center integrations

[Open](#)

Phone

Built by IBM - Plus

Give your system a voice so it can help customers over the phone.

[Learn more](#)

Contact center integrations

[Add](#)

Extensions

Add features or content to your assistant using custom or pre-built extensions.

[Learn more](#)

Build custom extension

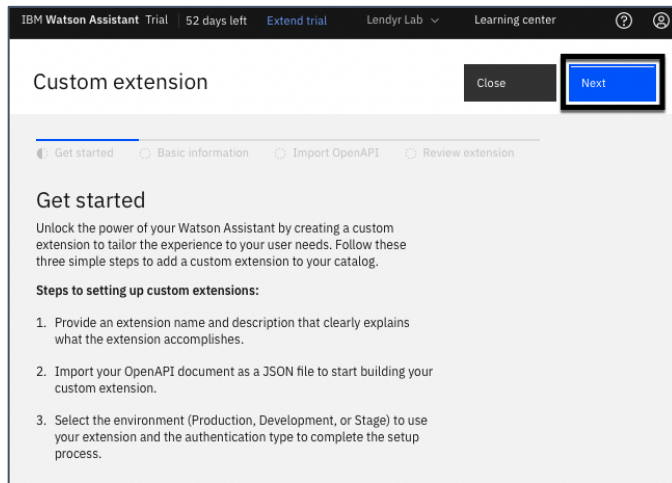
Search

Extend the scope of what your assistant can answer by searching through your documents and webpages.

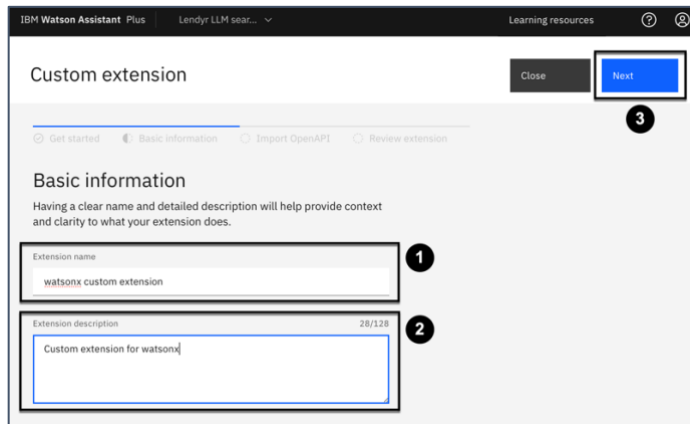
Segment

Get a better understanding of your users' end-to-end journeys by combining your assistant's data with other sources.

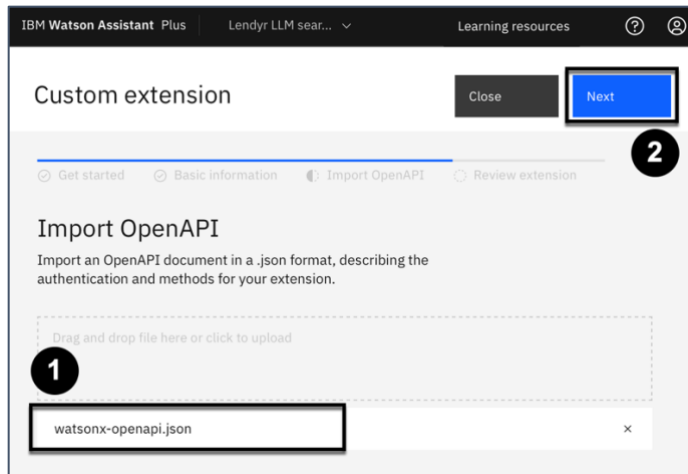
This first screen describes what you'll need to create the extension. Note that the watsonx OpenAPI JSON document, which you just obtained, is key to the setup. Click **Next** in the top right to proceed to the next screen:



The second screen asks you to name and describe the custom extension. Name it **watsonx custom extension (1)** and add a description, like “Custom extension for watsonx” (2). Click **Next (3)** to proceed to the next screen.

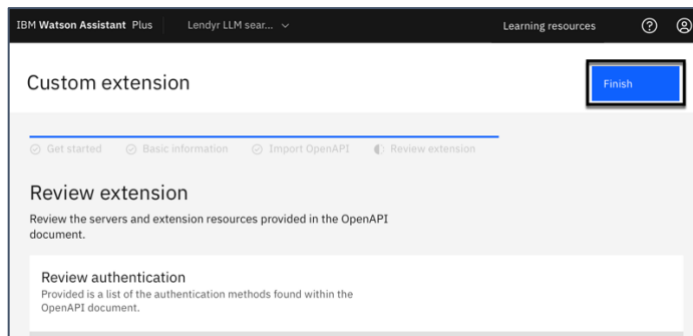


In the next screen, either drag-and-drop or click to upload the [watsonx OpenAPI specification file](#) (1) OpenAPI spec you downloaded, then click **Next (2)** to proceed to the next screen:

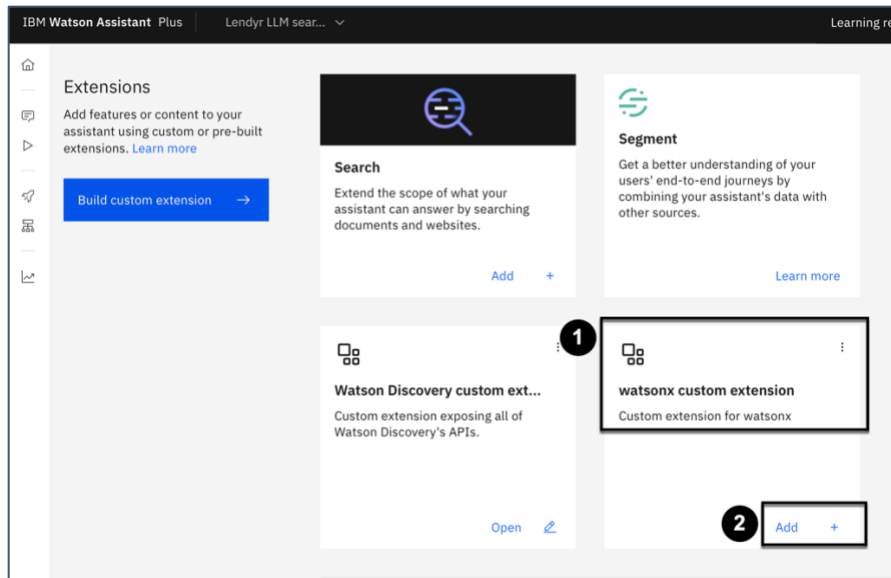


Note: If you get an error that the file is not in the proper JSON format, try downloading it again with another browser, and assure it has a **.json** extension.

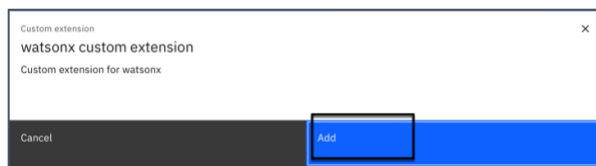
Click **Finish** to create the custom extension.



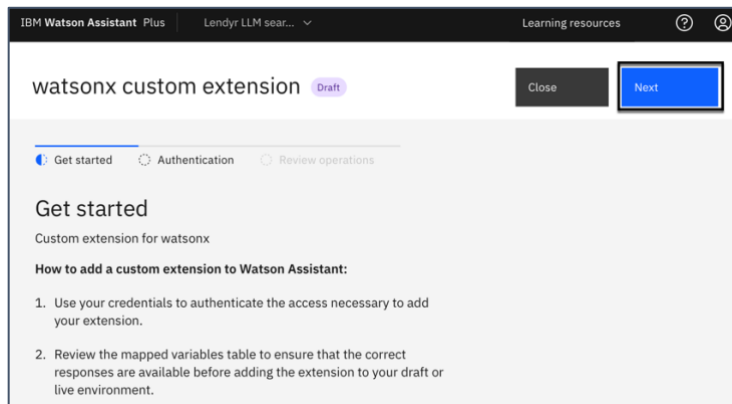
You should now be able to see the **watsonx custom extension (1)** in your Integrations catalog. Click **Add + (2)** so that you can configure a connection to your watsonx project:



Click **Add** on the popup screen.



Hit **Next** on the following screen:



On the Authentication screen,

1. Choose **OAuth 2.0** as the Authentication type.
2. Select **Custom apikey** as the Grant type.
3. Copy and paste the **IBM Cloud API key** you saved earlier into the Apikey field.
4. Click **Next**.

IBM Watson Assistant Plus

Lendyr LLM sear...

Learning resources

watsonx custom extension

Draft

Close

Next

Get started

Authentication

Review operations

Authentication

Authentication types are determined in the OpenAPI document and provide security for the extension.

Authentication type

OAuth 2.0

Grant type

Custom apikey

Custom Secrets

Apikey

ud17Q_LY(3zV6-FaQH10byubzryW9ChXvF10td;U2U

Client authentication

Send as Body

Header prefix

Bearer

Servers

https://us-south.ml.cloud.ibm.com - watsonx.ai v1-beta

Click **Finish**.

IBM Watson Assistant Plus

Lendyr LLM sear...

Learning resources

watsonx custom extension

Draft

Close

Finish

Get started

Authentication

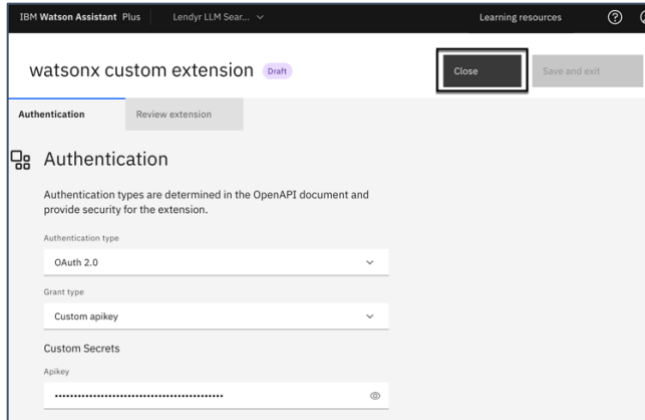
Review operations

Review operations

This table shows the operations defined in the OpenAPI document.

Operation	Method	Resource
Generation	POST	/ml/v1-beta/generation/text

Click **Close** on the final review screen.



Upload and configure the watsonx actions

Next, you will upload the actions your assistant will need, so download the [actions JSON file](#).

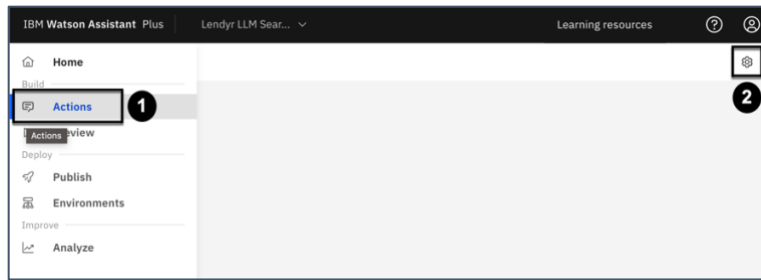
Notes: 1) You should not upload these actions directly into an *existing* assistant because doing so will overwrite your existing actions. 2) If the file downloads with an uppercase **.JSON** extension, you must rename it to lowercase **.json**, otherwise it will not upload correctly.

This file contains three actions:

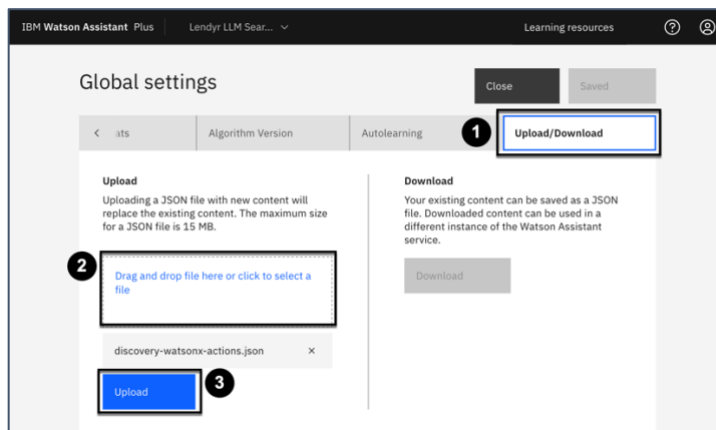
Action	Description
Search	Connects to Watson Discovery to search for documents related to the user query. The out-of-box "No Action Matches" action has been configured to send all input to this action, so whatever the user enters will be used as the search input. In turn, this action invokes the "Generate Answer" action to generate a response to the query.
Generate Answer	Configures the query prompt and document passages resulting from the Search action and calls the action "Invoke watsonx generation API." It is not meant to be invoked directly, but rather by the "Search" action.
Invoke watsonx generation API	Connects to watsonx and, using as context the documents resulting from the search, asks the language model to generate an answer to the user query. It is not meant to be invoked directly, but rather by the "Generate Answer" action.

The actions in this file will use search only when no action matches the user request. They search a complete watsonx project, and as such they are general-purpose and usable with any data set.

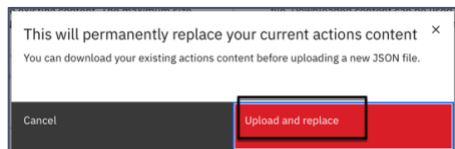
To upload the JSON file, click **Actions (1)**, and then **Global Settings (2)**:



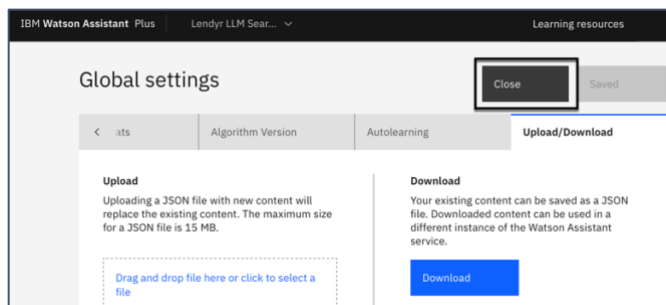
Scroll to the right until you are able to see and select the **Upload/Download (1)** tab. There, **drag and drop (2)** the [actions JSON file](#), and click **Upload (3)**:



Click **Upload and replace** at the warning pop-up screen.

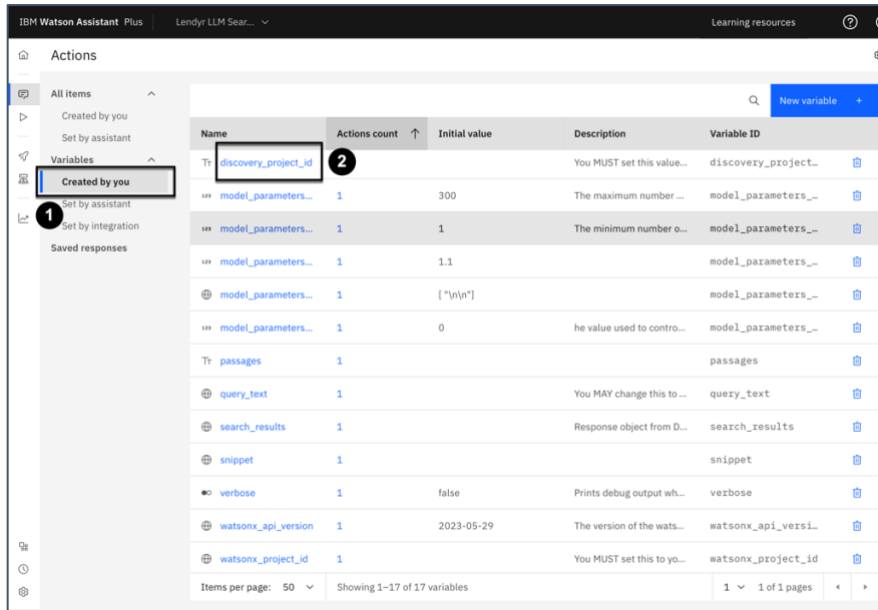


Hit **Close**.

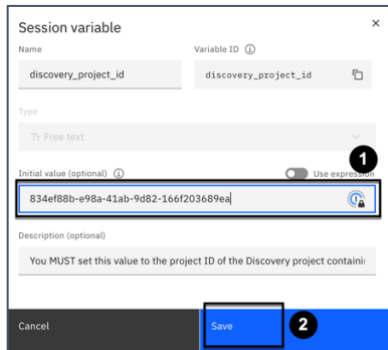


The actions JSON file you uploaded also includes variables used by the actions. You need to update two of them with your Discovery and watsonx project IDs.

Within the Actions page, under Variables, select the **Created by you (1)** filter, which displays the variables you just uploaded. Click the **discovery_project_id (2)** session variable:



Copy and paste the **project ID (1)** value you obtained when configuring Watson Discovery and click **Save (2)**.



Next, click the **watsonx_project_id** variable:

Name	Actions count	Initial value	Description	Variable ID
discovery_project_id	1		You MUST set this value...	discovery_project_...
model_parameters...	1	300	The maximum number ...	model_parameters_...
model_parameters...	1	1	The minimum number o...	model_parameters_...
model_parameters...	1	1.1		model_parameters_...
model_parameters...	1	["\n\n"]		model_parameters_...
model_parameters...	1	0	he value used to contro...	model_parameters_...
passages	1			passages
query_text	1		You MAY change this to ...	query_text
search_results	1		Response object from D...	search_results
snippet	1			snippet
verbose	1	false	Prints debug output wh...	verbose
watsonx_api_version	1	2023-05-29	The version of the wats...	watsonx_api_versi...
watsonx_project_id	1		You MUST set this to yo...	watsonx_project_id

Set the **watsonx_project_id (1)** variable to the watsonx project ID you obtained earlier and click **Save (2)**. This tells the assistant which watsonx project will be used for answer generation.

Session variable

Name: Variable ID:

Type:

Initial value (optional): ☐ Use expression

Description (optional):

Cancel

Those are all the variables you must set.

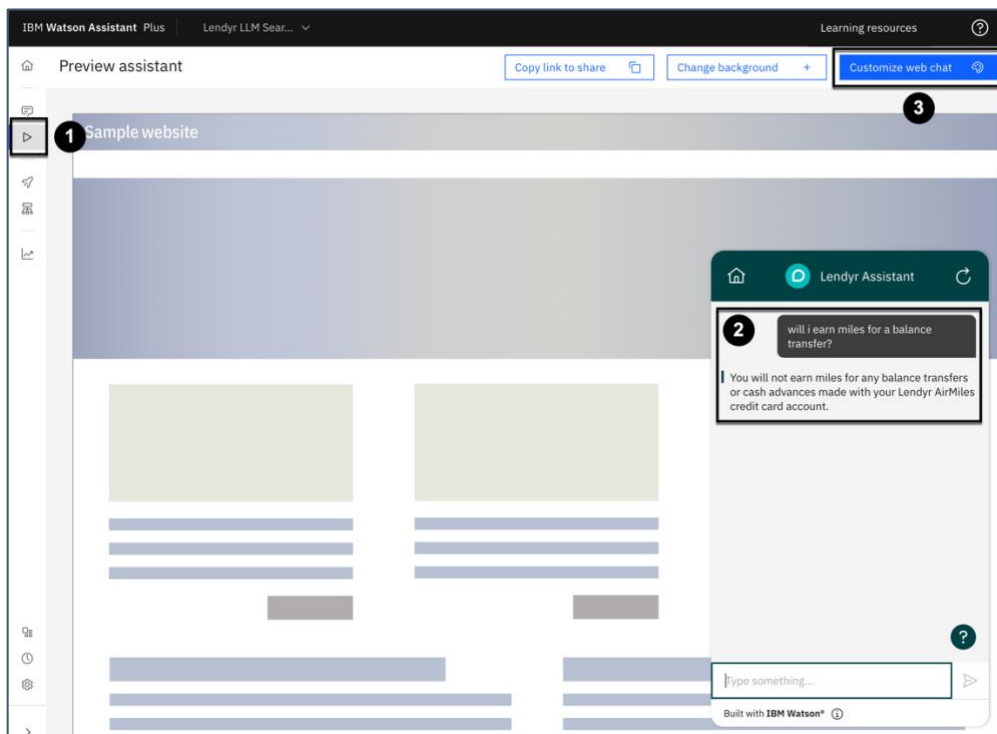
This list describes the available variables providing for greater control over your model:

- **discovery_date_version**: Discovery date versions are documented in the [release notes](#).
- **model_id**: The id of the watsonx model that you select for this action. Defaults to `google/flan-ul2`.
- **model_input**: The input to the watsonx model. You may change this to do prompt engineering, but a default will be used by the model if you don't pass a prompt here.
- **model_parameters_max_new_tokens**: The maximum number of new tokens to be generated. Defaults to 300.
- **model_parameters_min_new_tokens**: The minimum number of the new tokens to be generated. Defaults to 1.
- **model_parameters_repetition_penalty**: Represents the penalty for penalizing tokens that have already been generated or belong to the context. The range is 1.0 to 2.0 and defaults to 1.1.

- `model_parameters_stop_sequences`: Stop sequences are one or more strings which will cause the text generation to stop if/when they are produced as part of the output. Stop sequences encountered prior to the minimum number of tokens being generated will be ignored. The list may contain up to 6 strings. Defaults to ["\n\n"]
- `model_parameters_temperature` : The value used to control the next token probabilities. The range is from 0 to 1.00; 0 makes it deterministic.
- `model_response`: The text generated by the model in response to the model input.
- `passages`: Concatenation of top search results.
- `query_text`: You may change this to pass queries to Watson Discovery. By default the Search action passes the user's input.text directly.
- `search_results`: Response object from [Discovery query](#).
- `snippet`: Top results from the Watson Discovery document search.
- `verbose`: A boolean that will print debugging output if true. Default is false.
- `watsonx_api_version` - watsonx api date version. It currently defaults to 2023-05-29.
- `watsonx_project_id`: You must set this value to be [a project ID value from watsonx](#).

Try it out!

Now that you are finished configuring your assistant, try it out! Click on **Preview (1)** and enter the same question as before: **will I earn miles for a balance transfer (2)**. Note the clear, concise, and conversational answer. Compare this to the generic answer from ChatGPT and the excerpt answer provided by Watson Discovery earlier!

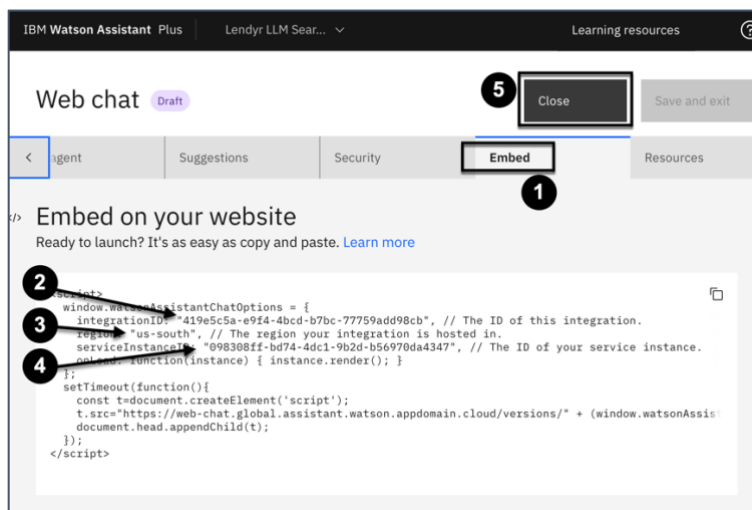


Preview your assistant on Lendyr Bank's website

You embedded your assistant on the Lendyr website in Section 2, so these directions may seem familiar.

On the **Preview** page (shown above), click **Customize web chat (3)**. Then, as shown below,

1. Click on the **Embed** tab.
2. Copy the value for your assistant's **integrationID** and paste it into a text document. You will use this value shortly.
3. Copy the value for your assistant's **region** and paste it into a text document. You will use this value shortly.
4. Copy the value for your assistant's **serviceInstanceId** and paste it into a text document. You will use this value shortly.
5. Finally, click **Close**.



Now, take your assistant's **integrationID**, **region**, and **serviceInstanceId**, and plug them into this URL:

[https://www.ibm.com/products/watson-assistant/demos/lendyr/demo.html?
integrationID=ID_HERE®ion=REGION_HERE&serviceInstanceId=ID_HERE](https://www.ibm.com/products/watson-assistant/demos/lendyr/demo.html?integrationID=ID_HERE®ion=REGION_HERE&serviceInstanceId=ID_HERE)

Make sure there are no spaces (each paste adds a space!), and then copy and paste your custom URL in your browser.

If you would like to demo the capabilities built in this section, review the [demo video](#) for a recommended flow with other utterances to ask the assistant.

Copy the new functionality to your Lendyr assistant

Note: There are two pre-requisites to being able to complete this last part of the lab. If both of these do not apply, simply skip to the last page of this document:

1. You must have completed sections 1 through 7 of the lab; and
2. Both assistants must be running in the same instance of watsonx Assistant (in other words, either under the same TechZone environment, or under the same instance under the same IBM Cloud user ID).

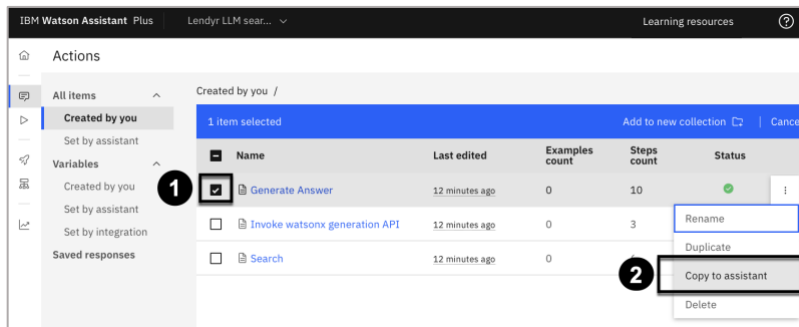
You can copy the LLM-powered search functionality you just created into your version of the Lendyr virtual assistant which you built in the previous lab sections.

Do not upload the actions to the assistant you previously built; this will overwrite your existing actions. Rather, you will [copy-and-paste](#) the actions.

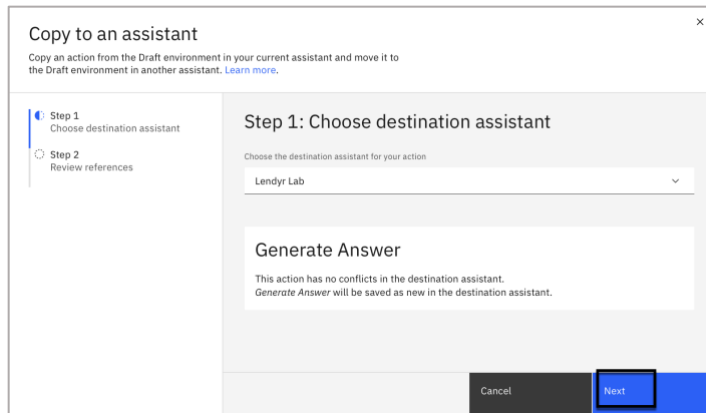
First, navigate to your Lendyr Lab assistant, built in sections 1-6. There, reconfigure the watsonx and Watson Discovery extensions by following the respective parts of this section:

- *Add the Watson Discovery extension*
- *Add the watsonx custom extension*

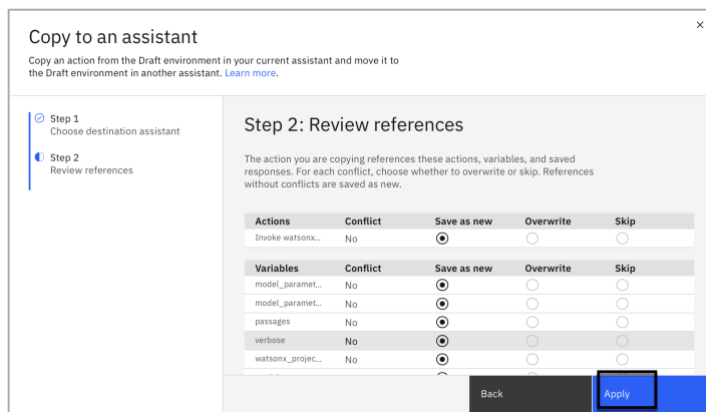
Then, navigate back to your new (LLM-powered) virtual assistant's **Actions** page. **Select (1)** each of the three actions, one at a time, and select [Copy to assistant](#) (2):



On the popup screen, check to see that you are pasting your Lendyr Lab assistant, and click **Next:**



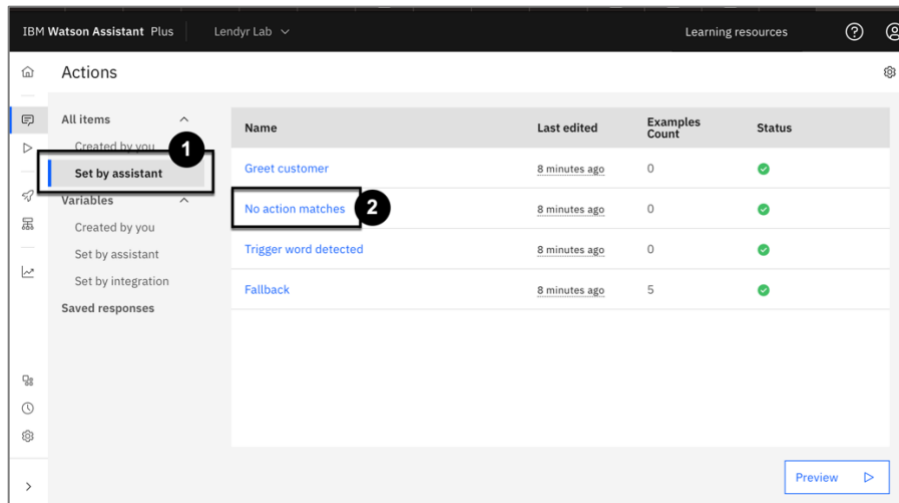
On the next screen, note that there are no “conflicts” and click **Apply**:



The virtual assistant may ask you to select *Overwrite* or *Save as new* for actions and variables. It will have default selections for each action and variable. Do not change the default selections.

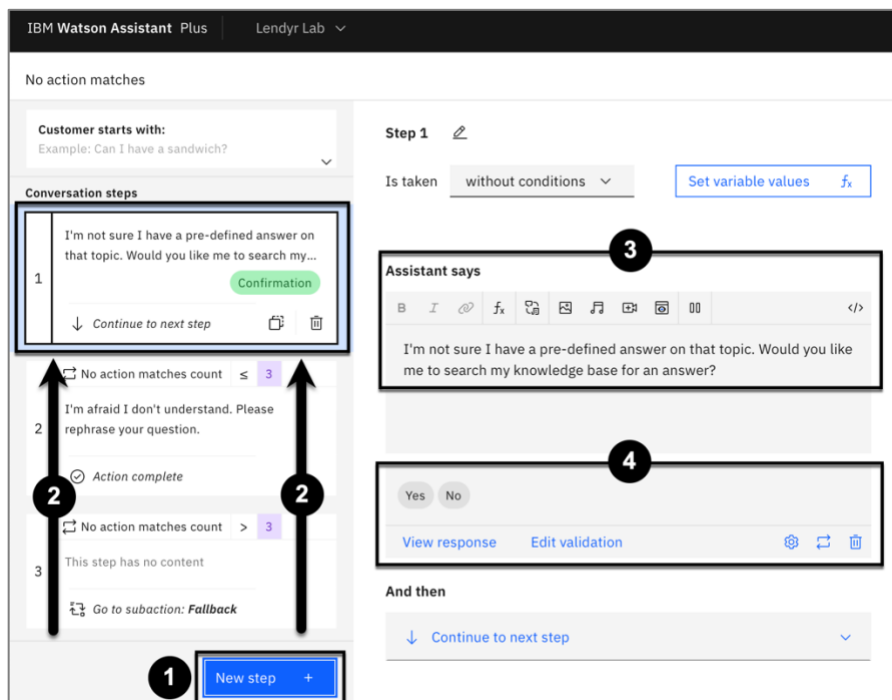
Now that you have copied the actions created in this (Section 7) of the lab to the Lendyr assistant you created in Sections 1-6, in your Lendyr virtual assistant, navigate to the **Actions** page.

Here, you will create an action that will ask the user if they want to try conversational search when the assistant does not recognize the question. To build this action click **Set by assistant (1)**, and then open the **No action matches (2)** action:



As shown below,

1. Add a **New step +**
2. Drag it up so it's the first step
3. Under **Assistant says**, write: "I'm not sure I have a pre-defined answer on that topic. Would you like me to search my knowledge base for an answer?"
4. **Define the customer response** as a Confirmation.

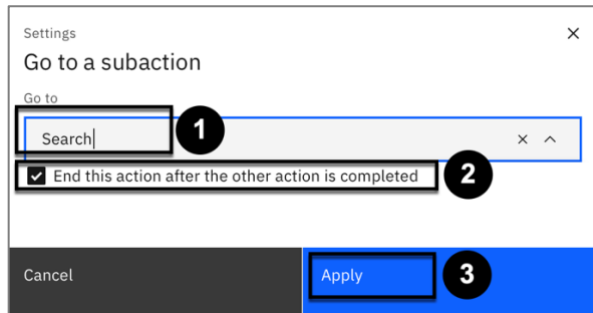


Next,

1. Add another **New step +**
2. Drag it up so it's the second step

3. Add the **condition**: If Step 1 is Yes
4. Under **Assistant says**, write: “OK!”
5. Change **And then** to **Go to a sub-action**

When the Go to a subaction search window appears, select **Search (1)** as the subaction, check the **End this action after the other action is completed (2)** checkbox, and click **Apply (3)**:



Finally,

1. Add another **New step +**
2. Drag it up so it's the third step
3. Add the **condition**: If Step 1 is No
4. Under **Assistant says**, write: “OK! In that case:”
5. **Save** the action.

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No action matches

Customer starts with:
Example: Can I have a sandwich?

Conversation steps

1 I'm not sure I have a pre-defined answer on that topic. Would you like me to... Confirmation
Continue to next step

1 is Yes
2 OK!
Go to subaction: Fallback

1 is No
3 OK! In that case:
Continue to next step

4 I'm afraid I don't understand. Please rephrase your question.
Action complete
No action matches count > 3
5 This step, [object Object], No action match count, Go to subaction: Fallback

Step 3

Is taken with conditions Set variable values f_x

Conditions 1 condition
If All of this is true:
1. I'm n... is No
and Add condition +
New condition group +

Assistant says
OK! In that case:
Define customer response
And then
Continue to next step

New step + Preview

Your Lendyr virtual assistant will now ask the end user if they want to try conversational search when the virtual assistant does not recognize their question.

Congratulations! You have now built the demo and functionality shown in [this video](#).

You have now completed this section of the lab, which also concludes this lab series. **You** have now covered **all** the fundamental concepts and features of watsonx Assistant. You are well-equipped to build and test a simple, powerful virtual assistant – with some practice, of course!

For IBMers and IBM business partners interested in digging deeper into watsonx Assistant, please complete one of these two courses, based on your role:

watsonx Assistant for Technical Sales Level 4	IBMers →	BPs →
watsonx Assistant for Practitioners Level 4	IBMers →	BPs* →

* Link coming soon

If you're unsure which is the right course for you, understand the course description and number of hours required to complete it, before making a selection.

IBM clients that are interested in deeper learning should discuss education options with their IBM or IBM business partner representative.

Report an issue

Report an issue, share feedback, or request a change or addition to the hands-on lab [here](#).